POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

A Lightweight Canonicalization Engine in C for Disciplined Convex Programming

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Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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Ce mémoire intitulé :

A Lightweight Canonicalization Engine in C for Disciplined Convex Programming

présenté par **William ZHANG**
en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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DEDICATION

To my parents, Xiaohong Zhang and Lingli He.

And to all the past, present and future `CVXPY` *contributors.*

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Throughout the last two years, I was incredibly fortunate to meet and work with many tremendous people, from students who have similar interests as me to experts in our field of optimization who were willing to share and discuss interesting problems. The work of my Master's thesis wouldn't have been possible without all the mentorship I received early on, as well as the collaboration opportunities I received later on. I would like to take this opportunity to give some regards to everyone who has been a part of this journey along with me.

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RÉSUMÉ

Les langages de modélisation pour l'optimisation convexe permettent d'écrire un problème sous une forme proche de sa formulation mathématique. Ils le compilent ensuite en données numériques acceptées par un solveur. Dans `CVXPY`, un langage de modélisation populaire, la majorité du temps de compilation sert à construire les matrices creuses du problème à partir d'arbres d'expressions symboliques. Les arbres expriment l'objectif et les contraintes en termes de fonctions des variables et des paramètres du problème. Les matrices creuses correspondent à la jacobienne de ces fonctions et à la hessienne de l'objectif. Leur construction revient donc à un calcul de dérivées creuses. Aucune des implémentations actuelles n'adopte ce point de vue. Pour les problèmes paramétriques, `CVXPY` met en cache une application linéaire des paramètres vers les données, coûteuse à construire. Pour tout autre problème, la compilation est répétée en entier à chaque résolution. Nous présentons `SparseDiffEngine`, une librairie C *open source* qui forme la jacobienne et la hessienne directement à partir de la structure des expressions. Le calcul est divisé en deux phases, une idée que nous empruntons aux interfaces des solveurs d'optimisation non linéaire. La phase symbolique s'exécute une seule fois par problème et détermine les structures creuses des matrices. La phase numérique remplit ensuite les valeurs dans ces structures fixes à chaque résolution. Pour un problème paramétrique, seule la phase numérique est répétée d'une résolution à l'autre. À l'aide de `SparseDiffEngine`, nous implémentons une nouvelle façon de former les données numériques dans `CVXPY`. Cette approche est une alternative à la mise en cache déjà offerte par `CVXPY` et ne limite pas la manière dont les paramètres apparaissent dans le problème. Nous comparons cette nouvelle approche aux trois implémentations actuelles de `CVXPY` sur une suite publique de 25 problèmes de référence. La première compilation est la plus rapide sur 21 des 25 problèmes, avec une accélération d'un facteur 2,8 en moyenne géométrique par rapport à la meilleure des trois. Résoudre de nouveau un problème paramétrique prend un temps comparable à celui de la mise en cache de `CVXPY`.

ABSTRACT

Modelling languages for convex optimization let users state problems in a form close to their mathematical formulation, and compile them into the numerical data that solvers accept. In `CVXPY`, a widely used modelling language, most of the compilation time is spent converting symbolic expression trees into the sparse constraint and objective matrices. The trees state the objective and constraints as functions of the variables and parameters. Because the constraint matrix is the Jacobian of these functions and the objective matrix is a Hessian, the conversion can be viewed as a sparse differentiation problem. None of the current implementations takes this perspective. For parametric problems, `CVXPY` caches a linear map from the parameters to the data, which is expensive to build. For every other problem, the compilation is repeated in full at every solve. We present `SparseDiffEngine`, a lightweight open-source C library that computes the Jacobian and the Hessian directly from the expression structure. The computation is divided into two phases, which is an idea we borrow from the interfaces of nonlinear programming solvers. A symbolic pass runs once per problem and determines the sparsity patterns of the matrices. A numeric pass fills new values into the fixed patterns at every solve. Re-solving a parametrized problem therefore costs only one numeric pass. Using `SparseDiffEngine`, we implement a new way to form the numerical data in `CVXPY`. This two-phase approach is an alternative to `CVXPY`'s existing caching of parametrized problems, and it places no restriction on how the parameters enter the problem. We compare this new approach with `CVXPY`'s three current implementations on a public benchmark suite of 25 problems. The first compilation is the fastest on 21 of the 25 problems, with a geometric-mean speedup of 2.8× over the best current implementation. Re-solves of parametrized problems run at times comparable to those of `CVXPY`'s existing caching.

TABLE OF CONTENTS

DEDICATION . . . iii

ACKNOWLEDGEMENTS . . . iv

RÉSUMÉ . . . vii

ABSTRACT . . . viii

LIST OF TABLES . . . xi

LIST OF FIGURES . . . xii

LIST OF SYMBOLS AND ACRONYMS . . . xv

CHAPTER 1 INTRODUCTION . . . 1
- 1.1 Problem statement . . . 2
- 1.2 Canonicalization . . . 4
  - 1.2.1 Conic reformulation . . . 5
  - 1.2.2 `CVXPY` . . . 6
  - 1.2.3 Example . . . 7
- 1.3 Computing derivatives . . . 9
  - 1.3.1 Numerical differentiation . . . 10
  - 1.3.2 Symbolic differentiation . . . 11
  - 1.3.3 Algorithmic differentiation . . . 11
- 1.4 Matrix computations . . . 12
  - 1.4.1 Dense matrices . . . 12
  - 1.4.2 Sparse matrices . . . 14
  - 1.4.3 Sparse matrix products . . . 15
- 1.5 Contributions . . . 16

CHAPTER 2 LITERATURE REVIEW . . . 17
- 2.1 Matrix stuffing in `CVXPY` . . . 17
  - 2.1.1 Composing linear operators . . . 18
  - 2.1.2 Example . . . 19
  - 2.1.3 Parameter caching . . . 20
- 2.2 Sparse derivatives from detected structure . . . 23
  - 2.2.1 Matrix-vector products and accumulation . . . 24

2.2.2 Compression by graph colouring . . . 25
2.2.3 `CasADi` . . . 26
2.3 Coefficient extraction from declared structure . . . 28
2.3.1 `AMPL` . . . 28
2.3.2 `JuMP` . . . 31
2.4 Research gap . . . 32

CHAPTER 3 A LIGHTWEIGHT CANONICALIZATION ENGINE . . . 34
3.1 Overview of the engine . . . 34
3.1.1 Atom classes . . . 35
3.1.2 Expression node . . . 36
3.2 Matrix representations . . . 36
3.2.1 Sparse matrix . . . 37
3.2.2 Permuted dense matrix . . . 37
3.2.3 Stacked permuted dense matrix . . . 39
3.3 Computational kernels . . . 39
3.3.1 Kernel dispatch . . . 40
3.3.2 Caching structure across solves . . . 41
3.4 Problem interface . . . 44
3.4.1 Stacking the constraint Jacobians . . . 44
3.4.2 `CVXPY` interface . . . 44
3.5 Summary . . . 46

CHAPTER 4 RESULTS . . . 48
4.1 First canonicalization . . . 49
4.2 Parametric canonicalization . . . 51
4.3 Discussion . . . 52

CHAPTER 5 CONCLUSION . . . 54
5.1 Limitations . . . 54
5.2 Future Research . . . 55

REFERENCES . . . 57

LIST OF TABLES

Table 1.1 A selection of `CVXPY` atoms with their DCP attributes: curvature, monotonicity in each argument, and sign of the output. 4

Table 2.1 A subset of the linear operators, classified as views or copies 19

Table 2.2 How the four modelling languages and `SparseDiffEngine` obtain and reuse the coefficients. 33

Table 3.1 The atom classes and how each computes its Jacobian. 35

Table 3.2 The kernels, the operation each computes, and the atoms of a canonicalized DCP problem that call them, grouped into unary and binary operations. In the second column, $\mathbf{A}$ denotes the Jacobian operand, $\mathbf{B}$ a data matrix, $\mathbf{d}$ a data vector, and $\mathbf{D} = \mathrm{diag}(\mathbf{d})$. 41

Table 4.1 Canonicalization time on all problems 51

Table 4.2 First canonicalization and re-solve on the parametric problems 52

LIST OF FIGURES

Figure 1.1 Example of an affine expression tree for $\mathbf{A}\mathbf{x} - \mathbf{b}$. Variables are drawn as circles and constants (or parameters) as boxes. 5

Figure 1.2 Before canonicalization. `CVXPY` represents the DCP problem (1.5) as these expression trees. Variables are labelled with a circle, and parameters with a square. 8

Figure 1.3 After canonicalization. `CVXPY` represents the conic form (1.6) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. The variable $\mathbf{x}$ appears as a leaf of several trees. All occurrences refer to the same variable. 10

Figure 1.4 The least squares objective $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$ with $\mathbf{A} \in \mathbb{R}^{2\times 2}$ as a scalar and as a vectorized computational graph. In the scalar graph every edge carries a scalar partial derivative. In the vectorized graph the local derivative of each node is a Jacobian block, shown below the node: $\mathbf{A}$ for the affine node and $2\mathbf{r}^\top$ for the sum of squares, where $\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$. 13

Figure 1.5 Dense storage of a 2×3 matrix: the same entries as one contiguous array in row-major and in column-major order. 13

Figure 1.6 The product $\mathbf{y} = \mathbf{S}\mathbf{v}$ with the matrix of (1.10). In CSR, the nonzeros of one row form a dot product that produces one output entry. In CSC, the nonzeros of one column are scaled by one input entry and accumulated into several output entries. 15

Figure 2.1 Jacobians under the vectorization (2.1), for a matrix variable $\mathbf{X}$ and the product $\mathbf{F}\mathbf{X}$, with $\mathbf{X}, \mathbf{F} \in \mathbb{R}^{2\times 2}$. The entries of $\mathrm{vec}(\mathbf{X})$, in column-major order, index the columns of the Jacobians. 18

Figure 2.2 The column offsets of a stack of three variables $\mathbf{z} = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ with $\mathbf{u} \in \mathbb{R}^2$, $\mathbf{v} \in \mathbb{R}^3$, and $\mathbf{w} \in \mathbb{R}^2$. Each offset is the total dimension of the preceding variables: $\mathbf{u}$ starts at column 0, $\mathbf{v}$ at 2, and $\mathbf{w}$ at 5. The entries of $\mathbf{z}$ index the columns of every augmented matrix $[\mathbf{A}_f \ \mathbf{b}_f]$. . . 20

Figure 2.3 Matrix stuffing of (1.5) by a bottom-up pass over the conic-form expression trees of Figure 1.3: each node's matrix is obtained by applying a linear operation to the matrices of its children. The coefficients appear negated because (1.2) writes the constraints as $\mathbf{b} - \mathbf{A}\mathbf{x} \in \mathcal{K}$. The horizontal rules in $[\mathbf{A}\ \mathbf{b}]$ separate the rows of the three constraints. . 21

Figure 2.4 What each layout stores during the pass, at the node $-(\mathbf{F}\mathbf{x} - \mathbf{g})$ of (1.5): a dictionary of narrow per-variable blocks, or a single matrix spanning all columns. 21

Figure 2.5 The DPP path on the objective data of (1.5): the parameter map $\mathbf{Q}$ is built and cached once, and each re-solve recovers $\mathbf{c}$ with the single sparse product $\mathbf{Q}\tilde{\boldsymbol{\theta}}$. 22

Figure 2.6 The `ignore_dpp` path on the objective of (1.5): the parameter leaf μ is replaced by a constant holding its current value 2.5, and the pass is repeated on the parameter-free tree on every change of the parameters. 22

Figure 2.7 Forward sweep of a computational graph 24

Figure 2.8 Reverse sweep of a computational graph 25

Figure 2.9 Accumulation by vertex elimination 25

Figure 2.10 Graph colouring compresses the seeds 27

Figure 3.1 The Jacobian of $\mathbf{B}\mathbf{x}$ with respect to the full variable vector, for dense data $\mathbf{B}$. Its nonzeros are exactly the entries of $\mathbf{B}$. 37

Figure 3.2 The Jacobian of $\mathbf{B}\mathbf{x}$ as a permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{4\times 2}$ and a variable $\mathbf{x} \in \mathbb{R}^2$ at columns $(4, 5)$ of a problem with ten variable entries. Only the dense block $\mathbf{X}$ is stored, and the selectors are the rows of the 4×4 and 10×10 identities indexed by $\mathbf{r}$ and $\mathbf{c}$. 38

Figure 3.3 Row indexing of the $\mathbf{A}$ of Figure 3.2. The result is mostly zeros, but its nonzeros still form one smaller dense block. 39

Figure 3.4 The Jacobian of $\mathbf{B}\mathbf{V}$ as a stacked permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{2\times 2}$ and a variable $\mathbf{V} \in \mathbb{R}^{2\times 2}$ at columns 4 to 7 of a problem with ten variable entries. There is one block per column of $\mathbf{V}$, and the 4×10 matrix $\mathbf{M}$ is never formed in practice. 40

Figure 3.5 Double dispatch of the transposed products $\mathbf{B}^\top\mathbf{A}$ and $\mathbf{B}^\top\mathbf{D}\mathbf{A}$. Each cell names the representation of the result for one combination of operand types (PD is permuted dense and sPD is stacked permuted dense), and its colour indicates how the product is computed. 42

Figure 3.6 The two passes of a sparse product $\mathbf{C} = \mathbf{AB}$. The symbolic pass finds, from the patterns of $\mathbf{A}$ and $\mathbf{B}$, the positions of $\mathbf{C}$ that can be nonzero, denoted by $*$, and stores them. The numeric pass computes one sparse dot product per stored position, over the indices at which the row of $\mathbf{A}$ and the column of $\mathbf{B}$ are both nonzero. 43

Figure 3.7 The two passes of the `sum` atom for the sum of an expression with three entries. The Jacobian of the sum is the sum of the three rows of the child Jacobian, whose nonzeros v_j are numbered row by row. The symbolic pass computes the pattern of the summed row and the map from each nonzero to the output entry w_i it is added to, shown as arrows. The numeric pass accumulates the values along the fixed arrows. 43

Figure 3.8 The Jacobians of three constraint trees stack into the global constraint Jacobian. The row pointer array `p` concatenates the pointers of the trees, shifted by the nonzeros of the blocks before them. The numeric pass copies the values of each tree into its slice of the value array `x`. The second tree is affine, so the copy is skipped after the first evaluation. 45

Figure 4.1 Performance profiles of canonicalization time and peak memory . . . 50

LIST OF SYMBOLS AND ACRONYMS

AD	algorithmic differentiation
AML	algebraic modelling language
AMPL	A Mathematical Programming Language
ASA	affine-solver-affine
ASL	AMPL Solver Library
BLAS	Basic Linear Algebra Subprograms
COO	coordinate
CSC	compressed sparse column
CSR	compressed sparse row
DAG	directed acyclic graph
DCP	disciplined convex programming
DPP	disciplined parameterized programming
JVP	Jacobian-vector product
LP	linear program
MOI	MathOptInterface
MPS	Mathematical Programming System
NLP	nonlinear program
QP	quadratic program
SIMD	single instruction, multiple data
SOCP	second-order cone program
SDP	semidefinite program
VJP	vector-Jacobian product

CHAPTER 1 INTRODUCTION

Since the early days of linear programming, optimization practitioners have needed to translate mathematical formulations into data that solvers can process. At first, this meant encoding each coefficient of the problem matrices by hand [1]. Matrix generators were introduced in the late 1960s to automate this task: they compute the coefficients through explicit loops and write them out in solver-readable formats like MPS [2, 3]. Matrix generators were able to remove most of the manual labour, but a gap still remained. Modellers naturally think in terms of sets, indices, and algebraic expressions, while procedural code constructs the coefficient matrices entry by entry. This mismatch between how a model is expressed and how it is implemented made models difficult to verify, modify, and communicate with others [4]. Algebraic modelling languages such as `AMPL` [5], `AIMMS` [6], and `GAMS` [7] closed this gap by accepting problem descriptions that closely follow the mathematical formulation. The survey [8] provides a complete overview of the early history of modelling languages.

More recently, modelling languages embedded in general-purpose programming languages have gained popularity. These domain-specific languages include `Pyomo` [9, 10] in Python, `JuMP` [11] in Julia, `CasADi` [12] in C++, and `YALMIP` [13] in MATLAB. Other domain-specific modelling languages specialized for convex optimization [14] are built on disciplined convex programming (DCP), a set of composition rules under which the convexity of a problem can be verified automatically [15, 16]. These DCP-based languages include `CVX` [17] in MATLAB, `CVXR` [18] in R, `CVXPY` [19] in Python, and `Convex.jl` [20] in Julia.

In the meantime, modern general-purpose solvers [21–23] have mostly transitioned towards conic programming, which generalizes linear programming and can represent nearly every convex program used in practice [24]. The translation task remains a responsibility of the modelling languages, which now target conic problem data instead of linear program (LP) coefficient matrices. In DCP-based languages, this translation process is called canonicalization [25]. While the time taken to solve an optimization problem is often dominated by the solver, the time taken to canonicalize (or compile) it can also be significant [26]. The main goal of this Master's thesis is to reduce the compilation time of `CVXPY`.

1.1 Problem statement

The standard form of a convex optimization problem is

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & f_0(\mathbf{x};\boldsymbol{\theta}) \\ \text{subject to} & f_i(\mathbf{x};\boldsymbol{\theta}) \leq 0, \quad i = 1,\ldots,m_1 \\ & g_j(\mathbf{x};\boldsymbol{\theta}) = 0, \quad j = 1,\ldots,m_2, \end{array} \tag{1.1}$$

where $\mathbf{x} \in \mathbb{R}^n$ is the optimization variable, the objective $f_0 : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ and inequality constraint functions $f_1, \ldots, f_{m_1} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ are convex, and the equality constraint functions $g_1, \ldots, g_{m_2} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ are affine [14]. The parameter $\boldsymbol{\theta} \in \mathbb{R}^p$ specifies data that can change between solves but is fixed and given when solving an instance of the problem. We call the parametrized problem (1.1) a problem family. For a fixed value of $\boldsymbol{\theta}$, we call it a problem instance.

Canonicalization converts the high-level description (1.1) to a quadratic conic form used by many general-purpose convex solvers

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & \frac{1}{2}\mathbf{x}^\top \mathbf{P}(\boldsymbol{\theta})\mathbf{x} + \mathbf{c}(\boldsymbol{\theta})^\top \mathbf{x} \\ \text{subject to} & \mathbf{b}(\boldsymbol{\theta}) - \mathbf{A}(\boldsymbol{\theta})\mathbf{x} \in \mathcal{K}, \end{array} \tag{1.2}$$

where the problem data are the symmetric positive semidefinite matrix $\mathbf{P}(\boldsymbol{\theta}) \in \mathbb{S}^n_+$, the matrix $\mathbf{A}(\boldsymbol{\theta}) \in \mathbb{R}^{m\times n}$, the vectors $\mathbf{b}(\boldsymbol{\theta}) \in \mathbb{R}^m$ and $\mathbf{c}(\boldsymbol{\theta}) \in \mathbb{R}^n$, and the non-empty closed convex cone $\mathcal{K} \subseteq \mathbb{R}^m$. Problem (1.2) generalizes both quadratic programs (QPs), when $\mathcal{K} = \mathbb{R}^m_+$ is the nonnegative orthant, and standard conic programs, when $\mathbf{P}(\boldsymbol{\theta}) = \mathbf{0}$.

Most of the cost of canonicalization lies in the final step, where symbolic expression trees are converted to numerical problem data. This step is often called matrix stuffing or the backend of canonicalization [27]. In `CVXPY`, the first solve of a problem family forms and caches a parameter tensor that maps the parameter vector to the problem data. On subsequent solves, new problem data can be obtained with a single matrix-vector product [26].

For large-scale problems, forming the parameter tensor can be prohibitively expensive, so `CVXPY` lets users opt out of it entirely. Users must thus choose between paying a large upfront cost on the first solve and repeating the full canonicalization, with nothing reused, at every solve. We review `CVXPY` and the frontend of canonicalization in §1.2, as well as the parameter tensor and matrix stuffing in §2.1.

The constraint matrix $\mathbf{A}$ is the Jacobian of the stacked constraint expressions and $\mathbf{P}$ is the Hessian of the objective, so matrix stuffing amounts to computing a sparse Jacobian and

Hessian. Each row of $\mathbf{A}$ holds the coefficients of one scalar constraint, and these coefficients are the partial derivatives of that constraint with respect to the entries of $\mathbf{x}$. Classical methods build these matrices column by column with multiple passes of algorithmic differentiation (AD), using graph colouring to group columns that can be recovered in a single pass [28, 29]. These methods treat the expression as a blackbox: the sparsity pattern must first be detected and the columns grouped before a single entry of the matrix is obtained. We review the basics of computing derivatives in §1.3 and present sparse AD and graph colouring methods in §2.2.

A modelling language, by contrast, knows which variables enter each constraint. `AMPL` (A Mathematical Programming Language) is split into two components: its commercial translator writes the structure of the model to a `.nl` file [30], and the open source AMPL Solver Library (ASL) evaluates constraints and their derivatives from this file inside the solver process [31]. `Pyomo` also interfaces to solvers through `.nl` files, among other formats.

Unlike `AMPL` and `Pyomo`, `JuMP` has no single standard form to target. It decouples modeller and solver through MathOptInterface (MOI), a typed in-memory interface [32]. The model is built constraint by constraint, and each constraint stores its coefficients directly, so extraction reduces to copying them into the format of the chosen solver. None of these approaches operates on expression trees: the AD tools record the expression as a flat tape of instructions, `AMPL` exchanges the model through a file, and `JuMP` stores lists of coefficients. In `CVXPY` the expression trees themselves are the input of the backend. We review `AMPL` and `JuMP` in §2.3.

Our approach. We make use of the fact that the sparsity patterns of $\mathbf{A}$ and $\mathbf{P}$ are fixed across a problem family, because the parameter $\boldsymbol{\theta}$ changes the values of the problem data but not their structure. We can therefore split matrix stuffing into a symbolic and a numeric phase. The symbolic phase runs once on the first solve, where it computes the sparsity patterns, allocates memory for all intermediate matrices, and caches other computations that remain valid across the problem family. The numeric phase then fills in new values at every solve. This separation between structure and values is inspired by the interfaces of nonlinear program (NLP) solvers such as `IPOPT` [33, 34].

The symbolic phase is much cheaper to form and cache than the parameter tensor, and the numeric phase is often similar in cost to applying the matrix-vector product. The problem data is also assembled entirely in memory rather than through a separate model file. Finally, we exploit structure within the sparsity patterns themselves, which often contain dense blocks of data. When possible, we store these blocks as dense matrices and compute with dense linear algebra kernels, rather than always relying on generic sparse formats and routines (we provide background on dense and sparse matrix computations in §1.4).

1.2 Canonicalization

The problem family (1.1) is written in functional form, meaning the objective and constraints are given as functions of the variables and parameters. In a modelling language, the optimization variables $\mathbf{x}_1, \ldots, \mathbf{x}_k$ are collected into a single vector $\mathbf{x} \in \mathbb{R}^n$, and likewise the parameters $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_l$ into $\boldsymbol{\theta} \in \mathbb{R}^p$ [14]. Users are not required to write constraints with a zero right-hand side as in (1.1). Modelling languages accept inequalities with a convex left-hand side and a concave right-hand side, and equalities with both sides affine. Converting to the form (1.1) is trivial and only requires moving every term to the left-hand side. Internally, each function is represented as an expression tree.

Modelling languages for convex optimization are built on top of a library of functions which can be used to describe problems. These functions are known as atoms in `CVXPY` because they are the fundamental building blocks of the modelling language [15]. Alongside its numerical definition, each atom declares useful attributes such as its curvature, monotonicity with respect to each argument, and the sign of its output. A selection of common atoms is shown in Table 1.1.

Table 1.1 A selection of `CVXPY` atoms with their DCP attributes: curvature, monotonicity in each argument, and sign of the output.

Atom	Definition	Curvature	Monotonicity	Sign
`sum`	$\sum_i x_i$	affine	non-decreasing	any
`abs`	$\lvert x \rvert$	convex	unknown	nonnegative
`square`	x^2	convex	unknown	nonnegative
`sqrt`	$\sqrt{x}$	concave	non-decreasing	nonnegative
`exp`	e^x	convex	non-decreasing	positive
`log`	$\log x$	concave	non-decreasing	any
`norm`	$\lVert x \rVert_2$	convex	unknown	nonnegative
`max`	$\max_i x_i$	convex	non-decreasing	any
`sum_largest`	sum of k largest entries	convex	non-decreasing	any

An expression is defined recursively as an atom evaluated at subexpressions, each of which is itself another expression. Such an expression is naturally stored as an expression tree, with atoms at its internal nodes and variables, parameters, and constants at its leaves. In Python, this tree can be formed automatically from user code through operator overloading. Most operations on an expression, from evaluation to verifying convexity to differentiation, traverse the tree and combine the intermediate results at each node.

To certify the convexity of a problem, DCP-based modelling languages use the DCP compo-

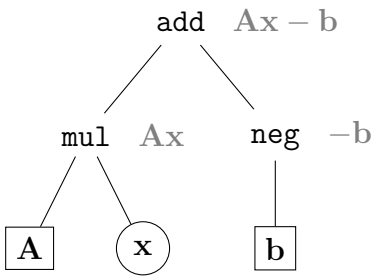


Figure 1.1 Example of an affine expression tree for $\mathbf{Ax} - \mathbf{b}$. Variables are drawn as circles and constants (or parameters) as boxes.

sition rules to classify the curvature of each function [15]. Once all functions are classified, a problem is DCP-compliant if the objective and each constraint have curvatures compatible with a convex problem: a convex objective, convex $\leq$ concave inequalities, and affine equalities. Otherwise, if a curvature is incompatible, for example when f_0 is concave, or cannot be determined, the modelling language raises an error and terminates. Because these rules are only sufficient for convexity, a problem that is genuinely convex but not expressed in a recognized form may still be rejected.

The composition $\phi(u_1(\mathbf{x}), u_2(\mathbf{x}), \ldots, u_k(\mathbf{x}))$ of an atom $\phi : \mathbb{R}^k \to \mathbb{R}$ with subexpressions $u_i : \mathbb{R}^n \to \mathbb{R}$ is convex when ϕ is convex and, for each argument $i \in \{1, \ldots, k\}$, one of the following holds:

- ϕ is non-decreasing in argument i and u_i is convex;
- ϕ is non-increasing in argument i and u_i is concave; or
- u_i is affine.

The same results hold for concave functions by swapping convex and concave. Affine expressions require ϕ and all of its arguments to be affine [15].

1.2.1 Conic reformulation

Once a problem is verified to be DCP-compliant, canonicalization rewrites its functional form (1.1) into an equivalent functional conic form

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & f_0(\mathbf{x}; \boldsymbol{\theta}) \\ \text{subject to} & h_i(\mathbf{x}; \boldsymbol{\theta}) \in \mathcal{K}_i, \quad i = 1, \ldots, m, \end{array} \tag{1.3}$$

where f_0 is affine or quadratic, each $h_i : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^{d_i}$ is affine, and each $\mathcal{K}_i \subseteq \mathbb{R}^{d_i}$ is one of the standard cones supported by general-purpose conic solvers: the zero, nonnegative, second-order, power, exponential, and positive semidefinite cones.

Every nonlinear atom is rewritten into conic constraints in three steps. The first step rewrites the problem into Smith form [35], where each nonlinear atom is replaced by a new variable coupled with an equality constraint. Every atom then acts only on affine arguments. For a convex atom ϕ, the Smith form gives an auxiliary variable $t \in \mathbb{R}$ with the equality constraint $\phi(\mathbf{x}) = t$. The second step relaxes this equality to the inequality $\phi(\mathbf{x}) \leq t$. The DCP rules allow a convex atom only where its value is minimized or upper-bounded, so the solver always prefers t as small as possible. The relaxation is therefore lossless. The pair $(\mathbf{x}, t)$ then lies in the epigraph $\{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \phi(\mathbf{x}) \leq t\}$ of ϕ, so we often refer to t as an epigraph variable. A concave atom relaxes to $\phi(\mathbf{x}) \geq t$ instead, giving a hypograph variable. The third and final step expresses each epigraph (hypograph) inequality as the membership of an affine function of $(\mathbf{x}, t)$ in a convex cone [36].

Example of reformulations. A first example is the ℓ_2-norm atom $\|\mathbf{x}\|_2$ of $\mathbf{x} \in \mathbb{R}^n$, whose epigraph $\|\mathbf{x}\|_2 \leq t$ is exactly the second-order cone constraint $(t, \mathbf{x}) \in \mathcal{K}_{\text{soc}}$, where $\mathcal{K}_{\text{soc}} = \{(t, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n : \|\mathbf{x}\|_2 \leq t\}$. A second example is the nondifferentiable absolute value atom $|x|$ of a scalar x, whose epigraph $|x| \leq t$ can be written as the pair of affine inequalities $-t \leq x \leq t$. A final example is the sum of the k largest entries of $\mathbf{x}$, which expands into a small LP: the epigraph $\texttt{sum_largest}(\mathbf{x}, k) \leq t$ holds if and only if there exist a scalar $q \in \mathbb{R}$ and a vector $\mathbf{w} \in \mathbb{R}^n$ such that

$$q\mathbf{1} + \mathbf{w} \geq \mathbf{x}, \qquad \mathbf{w} \geq \mathbf{0}, \qquad kq + \mathbf{1}^\top \mathbf{w} \leq t, \tag{1.4}$$

where the inequalities are applied element-wise. These conditions are obtained by linear programming duality [37], as the sum of the k largest entries is itself the optimal value of a small LP [14]. Here, q acts as a threshold and each w_i measures how far the entry x_i rises above it, so that the sum of the k largest entries is bounded by $kq + \mathbf{1}^\top \mathbf{w}$ [14].

1.2.2 `CVXPY`

`CVXPY` is a DCP-based modelling language embedded in Python [19], which began in 2013 as an implementation of the modelling principles of `CVX` [17]. It differed from `CVX` in three design decisions: modular components built around a few shared data structures, the expression tree formed by Python's operator overloading as the central representation, and support for parameters, which lets compilation work be reused when a problem is solved

repeatedly. These ideas matured with version 1.0, which redesigned canonicalization as a general reduction-based rewriting system [25]. The new design enabled the addition of other disciplined grammars [34, 38, 39], and the early parameter support grew into disciplined parameterized programming (DPP), a restricted grammar under which canonicalized programs can be updated efficiently [26].

The open source project has since grown through contributions from many other enthusiastic developers. A recent paper surveys the new canonicalization and modelling features introduced between versions 1.1 and 1.9 [27].

The rewriting from the functional form (1.1) to the functional conic form (1.3) is carried out as a sequence of reductions. Each reduction provides an `apply` step that converts a problem into an equivalent one, and a `retrieve` step that recovers a solution of the original problem from a solution of the rewritten one [25]. A first example is `FlipObjective`, which turns a maximization problem into a minimization one by flipping the sign of the objective, and flips it back in its `retrieve` step. A second example is `CvxAttr2Constr`, which replaces variable attributes such as nonnegativity with explicit constraints when the target solver does not support them natively.

The main reduction, `Dcp2Cone`, applies the epigraph reformulation of §1.2.1 in a bottom-up pass over the expression trees. The problem is first rewritten into Smith form, each resulting equality is relaxed to an epigraph (or hypograph) inequality, and each inequality is finally expressed as a cone membership [35, 36]. In practice the three steps are performed at once: each nonlinear atom is replaced directly by its epigraph variable and cone constraints, and the intermediate Smith form and its relaxation never appear explicitly.

1.2.3 Example

We consider the following simple fitting with regularization problem

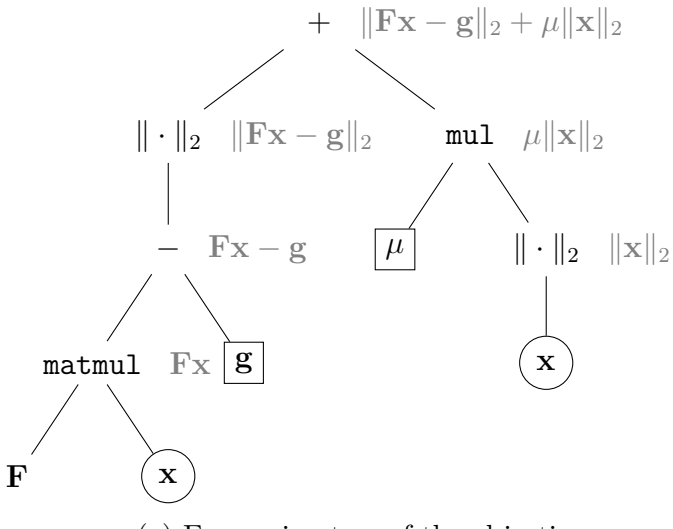
$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & \|\mathbf{F}\mathbf{x}-\mathbf{g}\|_2 + \mu\|\mathbf{x}\|_2 \\ \text{subject to} & \mathbf{x} \geq \mathbf{0}, \end{array} \tag{1.5}$$

with variable $\mathbf{x} \in \mathbb{R}^n$, fixed problem data $\mathbf{F} \in \mathbb{R}^{m\times n}$, and parameters $\mathbf{g} \in \mathbb{R}^m$ and $\mu \geq 0$. The constraint $\mathbf{x} \geq \mathbf{0}$ requires every entry of $\mathbf{x}$ to be nonnegative. Listing 1.1 states the problem in `CVXPY` code.

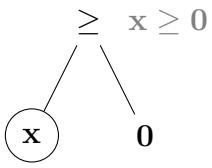
```
import cvxpy as cp
import numpy as np
F = np.random.randn(m, n)
x = cp.Variable(n)
g = cp.Parameter(m)
mu = cp.Parameter(nonneg=True)
objective = cp.Minimize(cp.norm(F @ x - g) + mu * cp.norm(x))
prob = cp.Problem(objective, [x >= 0])
```

Listing 1.1 The running example (1.5) in `CVXPY`.

The listing also uses the numerical computation library `NumPy` [40] for the data arrays. The expression tree representation of the problem is shown in Figure 1.2.



(a) Expression tree of the objective



(b) Expression tree of the constraints

Figure 1.2 Before canonicalization. `CVXPY` represents the DCP problem (1.5) as these expression trees. Variables are labelled with a circle, and parameters with a square.

After applying the reductions, the problem has been canonicalized to

$$\begin{array}{ll} \underset{t_1,t_2,\mathbf{x}}{\text{minimize}} & t_1+\mu t_2 \\ \text{subject to} & \mathbf{x}\geq\mathbf{0} \\ & (t_1,\mathbf{Fx}-\mathbf{g})\in\mathcal{K}_{\text{soc},m+1} \\ & (t_2,\mathbf{x})\in\mathcal{K}_{\text{soc},n+1}, \end{array} \tag{1.6}$$

where $t_1,t_2\in\mathbb{R}$ are new epigraph variables, and the cone $\mathcal{K}$ from (1.2) is the product

$\mathbb{R}^n_+ \times \mathcal{K}_{\text{soc},m+1} \times \mathcal{K}_{\text{soc},n+1}$. Problem (1.6) is now in the functional conic form of §1.2.1: every constraint is a cone membership, and the objective and constraint expressions are affine. Comparing the DCP trees of Figure 1.2 with the trees of Figure 1.3 shows the effect of canonicalization: the nonlinear norm atoms have been replaced by epigraph variables, leaving affine expression trees in the variables, which we call the conic-form trees of the problem. The three rows of (1.6) remain three separate cone constraints, each carrying the affine expressions that must lie in its cone. Figure 1.3 accordingly shows one expression tree per constraint, rather than a single combined tree.

1.3 Computing derivatives

Computing derivatives is a fundamental task in scientific computing, with many methods that differ in cost, in accuracy, and in the form of the result [41]. The derivatives in this Master's thesis are the constraint Jacobian $\mathbf{A}$ and the objective Hessian $\mathbf{P}$ of a canonicalized problem, both large sparse matrices. This section presents the Jacobian and the Hessian, the elementary methods for computing them, and the choices that distinguish the software implementations. §2.2 continues from these basics and reviews sparse AD and graph colouring methods.

For a vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ of an input $\mathbf{x} \in \mathbb{R}^n$, the first-order derivative is the Jacobian $\mathbf{J} \in \mathbb{R}^{m \times n}$ of partial derivatives of the outputs with respect to the inputs,

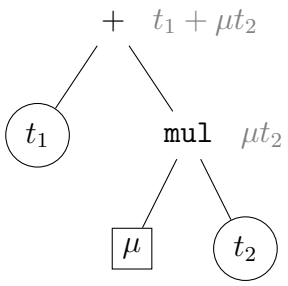
$$J_{ij} = \frac{\partial F_i}{\partial x_j}, \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n. \tag{1.7}$$

We mainly consider the constraint function, which stacks the potentially many affine expressions of a canonicalized problem into a single vector function of the variables. Its Jacobian is the constant constraint matrix $\mathbf{A}$ of (1.2), whose sparsity pattern is fixed although its values may change from one solve to the next.

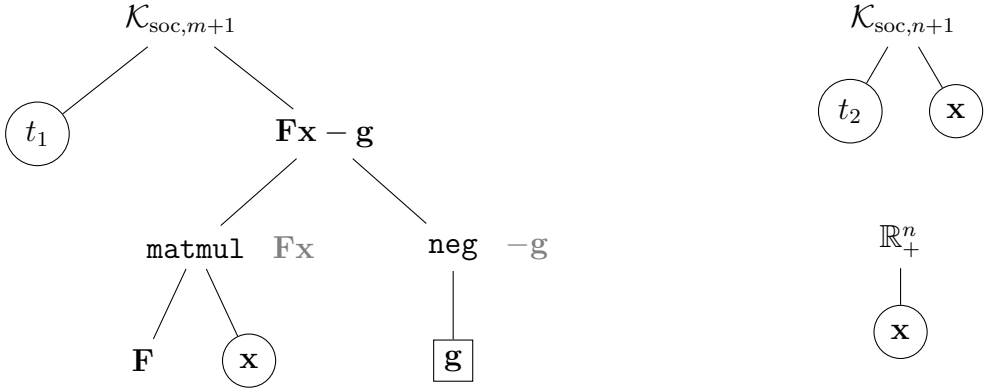
For a scalar function $f : \mathbb{R}^n \to \mathbb{R}$, the second-order derivative is the Hessian $\mathbf{H} \in \mathbb{R}^{n \times n}$ of second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}, \qquad i, j = 1, \ldots, n. \tag{1.8}$$

The Hessian is symmetric whenever f is twice continuously differentiable, and it is then the Jacobian of the gradient of f. It can therefore be obtained by differentiating the gradient with the same methods used for the Jacobian, although dedicated methods that compute it directly also exist. The function we consider is the canonicalized objective, whose Hessian is



(a) Expression tree of the objective



(b) One expression tree per constraint

Figure 1.3 After canonicalization. `CVXPY` represents the conic form (1.6) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. The variable $\mathbf{x}$ appears as a leaf of several trees. All occurrences refer to the same variable.

the matrix $\mathbf{P}$ of (1.2).

1.3.1 Numerical differentiation

Other than computing derivatives analytically, the simplest method is to use numerical differentiation, which approximates the derivative of a function at a point by evaluating the function at nearby points using a finite difference formula such as

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{1.9}$$

where h is a small step size. The central variant $f'(x) \approx (f(x+h) - f(x-h))/2h$ also uses two function evaluations, but the forward difference can reuse the already computed value $f(x)$ across directions, while the central variant cannot. The approach is simple but

has two downsides. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to truncation errors, when h is too large, or round-off errors, when h is too small, in the computed derivative if the step size is not carefully chosen [42]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations, also known as a derivative checker [43].

1.3.2 Symbolic differentiation

Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This symbolic approach to differentiation is used in computer algebra systems such as `Mathematica` [44] and `Maple` [45] as well as the `SymPy` library in Python [46]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can quickly form exponentially large symbolic expressions, which is impractical and inefficient for many problems [47]. When intermediate expressions are stored as common subexpressions, and the expression is represented with a directed acyclic graph (DAG) instead of a tree, symbolic differentiation performs the same set of operations as reverse mode differentiation [48].

1.3.3 Algorithmic differentiation

Algorithmic differentiation (AD) treats the function as a computational graph of elementary operations, each with a known local derivative rule. The chain rule is then applied to combine these local derivatives throughout the graph, giving the derivative of the whole function. AD is also commonly referred to as automatic differentiation. We prefer the term algorithmic to emphasize that AD differentiates the algorithm that computes a function, rather than a numerical approximation or a symbolic formula [47].

In this broad sense, AD can refer to any method that applies the chain rule over a computational graph. Its implementations differ in four basic choices: whether the graph is scalar or vectorized, whether the chain rule runs forward or in reverse, whether the derivative is assembled directly or accessed through matrix-vector products, and whether the result is stored sparse or dense.

In most AD tools, every node is a single elementary operation, an addition, a multiplication, or a univariate function, and every edge carries a scalar partial derivative [47]. In a vectorized graph, a node is a whole vector or matrix operation, and its local derivative is a matrix, a block of the Jacobian. Figure 1.4 contrasts the two on a small least squares objective. `CasADi` [12] offers both, through its scalar SX and vectorized MX expressions.

The modes of AD differ in the order in which the local derivatives are accumulated. Forward mode (or tangent mode) propagates derivatives from the inputs toward the outputs, following the order of evaluation. Reverse mode (or adjoint mode) propagates them from the outputs back toward the inputs, and requires storing the intermediate values of the evaluation [41]. In forward mode the work scales with the number of input variables, and in reverse mode with the number of output variables.

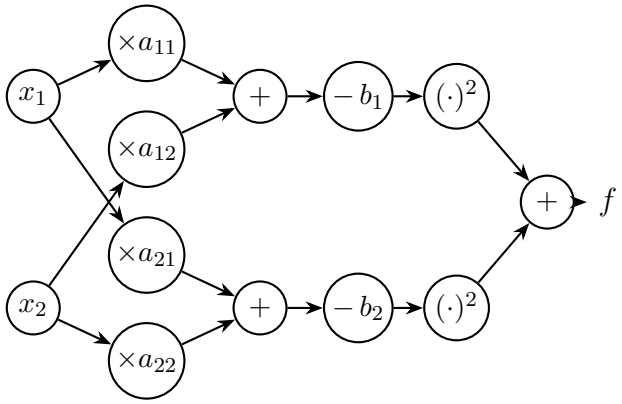
1.4 Matrix computations

The derivatives of §1.3 are matrices, and every matrix is stored in either a dense representation, which keeps all entries in a single array, or a sparse representation, which keeps only the nonzero entries alongside their positions. The representation determines the memory a matrix occupies and the algorithms that operate on it, and each has its own use cases. This section reviews the storage formats, algorithms, and some of the history behind both types of matrices.

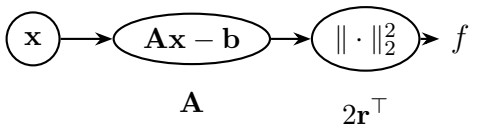
1.4.1 Dense matrices

A dense matrix is stored as a single contiguous array of its entries, ordered row by row (row-major) or column by column (column-major), as shown in Figure 1.5. The position of an entry is computed directly from its indices, so no index arrays are needed, and rows or columns are read sequentially in memory. C arrays are stored in row-major order and Fortran arrays in column-major order, so we often refer to the two orders as C-style and Fortran-style. Most dense linear algebra interfaces follow the Fortran convention.

Dense linear algebra is standardized by the Basic Linear Algebra Subprograms (BLAS), an interface of vector and matrix kernels [49, 50]. The BLAS is a specification rather than an implementation. Implementations such as `OpenBLAS`, Intel `MKL`, and Apple `Accelerate` are specialized for their respective processors, and nearly every machine has one available, whether shipped with the operating system or bundled by numerical libraries. `LAPACK`, the library behind dense factorizations and linear system solving in most scientific software, is built on top of the BLAS kernels [51].



(a) Scalar graph: one node per elementary operation



(b) Vectorized graph: one node per matrix operation

Figure 1.4 The least squares objective $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|_2^2$ with $\mathbf{A} \in \mathbb{R}^{2\times 2}$ as a scalar and as a vectorized computational graph. In the scalar graph every edge carries a scalar partial derivative. In the vectorized graph the local derivative of each node is a Jacobian block, shown below the node: $\mathbf{A}$ for the affine node and $2\mathbf{r}^\top$ for the sum of squares, where $\mathbf{r} = \mathbf{Ax} - \mathbf{b}$.

$$\mathbf{D} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

row-major (C)

1	2	3	4	5	6

column-major (Fortran)

1	4	2	5	3	6

Figure 1.5 Dense storage of a 2×3 matrix: the same entries as one contiguous array in row-major and in column-major order.

The BLAS interface is organized in three levels by the ratio of computation to data. Level 1 covers vector operations such as dot products, scaled copies, and additions, level 2 covers matrix-vector products, and level 3 covers matrix-matrix products. The routine names abbreviate the precision and the operation. For example, `dgemv` is the double-precision general matrix-vector product, and `dgemm` is the double-precision general matrix-matrix product.

BLAS implementations are fast because they are organized around the memory hierarchy. A modern processor multiplies numbers far faster than it loads them from main memory, so optimized kernels reuse data already in cache and process several entries per instruction with single instruction, multiple data (SIMD) units [52]. Level-3 kernels benefit the most because they perform a cubic number of operations on a quadratic amount of data, so each entry loaded from memory is reused many times.

1.4.2 Sparse matrices

In optimization problems, the matrices $\mathbf{P}$ and $\mathbf{A}$ often contain mostly zero entries. Such matrices are called sparse, and conic solvers expect their problem data in sparse formats. Techniques that store and compute with only the nonzero entries date back to the linear programming computations of the 1950s and the work of Markowitz [53]. The simplest sparse matrix data structure is the coordinate (COO) format, also called the triplet format. It consists of two integer index arrays `rows` and `cols` and a value array `vals` of length equal to the number of nonzero entries in the matrix. For example, the matrix

$$\mathbf{S} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 3 & 4 \\ 0 & 5 & 6 & 0 \end{bmatrix}, \tag{1.10}$$

is stored in COO format as in Listing 1.2, where we use zero-based indexing and array declaration notation from the C programming language [54].

```
int    rows[] = {0, 1, 1,  1,  2,  2};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

Listing 1.2 The matrix (1.10) in COO format.

The COO format is the easiest to understand and form, but it is not effective for most operations on sparse matrices. The compressed sparse row (CSR) format is commonly used instead.

The CSR format differs from the COO format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, Listing 1.3 represents the matrix (1.10) in CSR format, where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays.

```
int  idxptr[] = {0, 1,         4,      6};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

Listing 1.3 The matrix (1.10) in CSR format.

The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very cache-friendly because the nonzero entries are stored contiguously in memory. For this reason, the

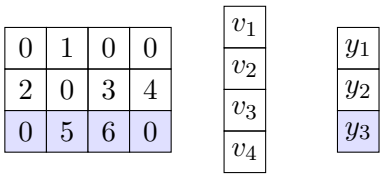
CSR format is the default for iterative methods to solve sparse linear systems [55].

The compressed sparse column (CSC) format applies the same compression to the columns instead. The pointer array `idxptr` now points into a `rows` array, and the nonzero entries are stored column by column. The CSC arrays of a matrix read exactly as the CSR arrays of its transpose, so transposition is a conversion between the two formats, computable without sorting in one counting pass over the nonzeros [56]. Accessing a column of a CSR matrix instead requires a scan of every row, so algorithms that read columns prefer CSC. Direct methods for sparse linear systems are the standard example [57].

The COO, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [58] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

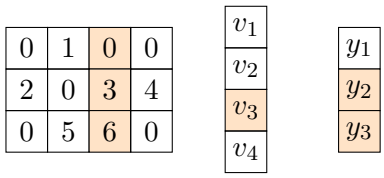
1.4.3 Sparse matrix products

Both compressed formats multiply a dense vector in one pass over the nonzeros, but the loop must follow the storage. CSR gathers, computing each output entry as a sparse dot product along its row, while CSC scatters, accumulating each column's contribution into the output vector. Figure 1.6 traces one product of the matrix in (1.10) under both formats.



$y_3 = 5v_2 + 6v_3$

(a) CSR gathers along a row



$y_2 \mathrel{+}= 3v_3, \quad y_3 \mathrel{+}= 6v_3$

(b) CSC scatters a column

Figure 1.6 The product $\mathbf{y} = \mathbf{S}\mathbf{v}$ with the matrix of (1.10). In CSR, the nonzeros of one row form a dot product that produces one output entry. In CSC, the nonzeros of one column are scaled by one input entry and accumulated into several output entries.

The product of two sparse matrices is formed row by row with Gustavson's algorithm [56]. Each row of the result is a combination of rows of the second operand, weighted by the nonzeros of the corresponding row of the first. Because the sparsity pattern of the result is not known in advance, the row is summed in a dense work array and then written back to sparse form [59]. It is the standard sparse matrix product, implemented in popular packages such as `Eigen` [60] and `scipy.sparse`.

In contrast to the BLAS kernels of §1.4.1, sparse products access memory irregularly. The loops run over the index arrays, so every nonzero costs an index load and an indirect access. Contiguous reads, SIMD, and cache tiling do not apply. Sparse formats are also a poor fit for dense data. A dense block stored in COO format triples its storage, one index pair per entry, and multiplying it through the sparse loop is far slower than the equivalent BLAS call. A sparse BLAS standard was specified by the BLAS Technical Forum [61], but implementations are few and far less widely adopted than the dense BLAS.

1.5 Contributions

In this Master's thesis, we make the following contributions:

1. We build `SparseDiffEngine`, a C library with Python bindings that assembles sparse Jacobians and Hessians directly from the expression structure, with no sparsity detection or graph colouring.
2. We implement a canonicalization backend for `CVXPY` built on `SparseDiffEngine`. This backend avoids the large upfront cost of the parameter tensor while still caching computations for subsequent solves.
3. We benchmark `SparseDiffEngine` against `CVXPY`'s existing canonicalization backends.

All code contributions are open source and available at

`https://github.com/SparseDifferentiation/SparseDiffEngine`
`https://github.com/SparseDifferentiation/Jacobian-accumulation`

The rest of this thesis is organized as follows. Chapter 2 reviews how `CVXPY` and other modelling languages compute the problem data. In Chapter 3, we present an overview of the main data structures and algorithms in `SparseDiffEngine` as well as its interface in `CVXPY`. Finally, Chapter 4 benchmarks this engine against `CVXPY`'s three canonicalization backends, and we conclude with a summary of our contributions and future improvements in Chapter 5.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the implementation of `CVXPY`'s matrix stuffing and of other modelling language approaches, from sparse AD in `CasADi` to coefficient extraction in `AMPL` and `JuMP`. These modelling languages all produce the same output as `CVXPY`, the solver problem data, but differ in the inputs they accept and in how they compute the coefficients. For example, `CasADi` accepts vectorized expressions as input, but differentiates them through matrix-vector products, so the structure of the derivatives must be detected at runtime, over many sweeps of the expression graph. `AMPL` and `JuMP` let the user declare vectorized expressions but scalarize them when processing the data into coefficients. The canonicalization backends of `CVXPY`, in contrast, keep the expressions vectorized and need no detection step. These differences are summarized in §2.4.

2.1 Matrix stuffing in `CVXPY`

Matrix stuffing converts the conic-form expression trees into the problem data. After canonicalization every expression is affine, so the Jacobian of each node is a constant matrix, which matrix calculus [62] gives in closed form. Computing the problem data reduces to composing these constant Jacobians over the tree. This section reviews the derivative rules along with two design choices of the `CVXPY` implementation: coefficient storage during the pass, and caching of the problem data across parameter changes. Finally, we describe `CVXPY`'s current backends and their limitations.

Matrix calculus extends differentiation to functions whose arguments and values are vectors and matrices. Systematic rules for such derivatives date back to the 1940s [63]. The simplest convention is to vectorize [62]: every expression is identified with

$$\operatorname{vec}(\mathbf{X}) = \begin{bmatrix} x_{11} & \cdots & x_{m1} & x_{12} & \cdots & x_{mn} \end{bmatrix}^\top, \tag{2.1}$$

the vector that stacks the columns of $\mathbf{X} \in \mathbb{R}^{m \times n}$, and the derivative of a vectorized expression with respect to another is an ordinary Jacobian matrix. Under this convention, matrix operations become linear operators on the vectorized arguments, and their Jacobians are structured. For example, the matrix product $\mathbf{FX}$ has the Kronecker-structured, block-diagonal Jacobian $\mathbf{I} \otimes \mathbf{F}$, shown in Figure 2.1.

$$\texttt{X = cp.Variable((2, 2))} \quad \Longrightarrow \quad \frac{\partial\, \mathrm{vec}(\mathbf{X})}{\partial\, \mathrm{vec}(\mathbf{X})} = \begin{array}{c} \\ x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

(a) A matrix variable is the identity map of its own entries.

$$\texttt{Y = F @ X} \quad \Longrightarrow \quad \frac{\partial\, \mathrm{vec}(\mathbf{Y})}{\partial\, \mathrm{vec}(\mathbf{X})} = \mathbf{I} \otimes \mathbf{F} = \begin{array}{c} \\ y_{11} \\ y_{21} \\ y_{12} \\ y_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} f_{11} & f_{12} & 0 & 0 \\ f_{21} & f_{22} & 0 & 0 \\ 0 & 0 & f_{11} & f_{12} \\ 0 & 0 & f_{21} & f_{22} \end{bmatrix} \end{array}$$

(b) The Jacobian of the product $\mathbf{Y} = \mathbf{FX}$ is block-diagonal, one copy of $\mathbf{F}$ per column of $\mathbf{X}$.

Figure 2.1 Jacobians under the vectorization (2.1), for a matrix variable $\mathbf{X}$ and the product $\mathbf{FX}$, with $\mathbf{X}, \mathbf{F} \in \mathbb{R}^{2\times 2}$. The entries of $\mathrm{vec}(\mathbf{X})$, in column-major order, index the columns of the Jacobians.

2.1.1 Composing linear operators

Every node of a conic-form tree, not only the root, computes an affine function of the problem variables and can be written in the vectorized form

$$f(\mathbf{z}) = \mathbf{A}_f\, \mathbf{z} + \mathbf{b}_f = \begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix}, \tag{2.2}$$

where $\mathbf{z}$ is the stacked vector of all the problem variables, $\mathbf{A}_f$ is the Jacobian of f with respect to $\mathbf{z}$, and $\mathbf{b}_f$ is a constant vector. Stacking the augmented matrices of the constraint trees yields the problem data $[\mathbf{A}\ \mathbf{b}]$ of (1.2). Matrix stuffing computes the augmented matrix in one bottom-up pass over the tree. A variable leaf contributes an identity block and a constant leaf contributes its value to the last column, as shown in Figure 2.3. An interior node with a single argument applies a constant linear operator $\mathbf{L}$ to the argument, which has the affine form (2.2) with augmented matrix $[\mathbf{A}_1\ \mathbf{b}_1]$. Applying $\mathbf{L}$ gives

$$\begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} = \mathbf{L} \begin{bmatrix} \mathbf{A}_1 & \mathbf{b}_1 \end{bmatrix}. \tag{2.3}$$

The operator $\mathbf{L}$ is the Jacobian of the node with respect to its argument, so (2.3) is exactly the chain rule of matrix calculus. A node with k arguments applies one operator per argument,

$f = \sum_{i=1}^{k} \mathbf{L}_i f_i$, which gives

$$\begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} = \sum_{i=1}^{k} \mathbf{L}_i \begin{bmatrix} \mathbf{A}_i & \mathbf{b}_i \end{bmatrix}, \tag{2.4}$$

so the node costs one matrix product per argument.

In `CVXPY`, the linear operators divide into views and copies [64]. A view performs a simple manipulation of the coefficients of its argument. Atoms with one constant argument, such as `matmul` and `multiply`, make copies instead. The operator is a constant matrix built from the constant argument, and its product with the augmented matrix of the other argument is stored as a new coefficient matrix. Table 2.1 classifies a subset of the operators.

Table 2.1 A subset of the linear operators, classified as views or copies, applied to an argument f_1 with augmented matrix $[\mathbf{A}_1 \ \mathbf{b}_1]$, where $\otimes$ denotes the Kronecker product, $\odot$ the elementwise product, / the elementwise division, $\mathcal{I}$ an index set, and $\operatorname{diag}(\mathbf{v})$ the diagonal matrix with the entries of $\mathbf{v}$ on its diagonal.

Atom	Operator $\mathbf{L}$	Action on $[\mathbf{A}_1 \ \mathbf{b}_1]$
Views		
$-f_1$	$-\mathbf{I}$	flip the sign of every stored entry
$f_1^\top$	$\mathbf{K}$, the vec-permutation	permute the rows
$f_1[\mathcal{I}]$	$\mathbf{P}_\mathcal{I}$, a row selection	keep the rows indexed by $\mathcal{I}$
$\operatorname{reshape}(f_1)$	$\mathbf{I}$	none, the vectorization is unchanged
Copies		
$\mathbf{F} f_1$	$\mathbf{I} \otimes \mathbf{F}$	multiply by a block diagonal formed from $\mathbf{F}$
$f_1 \mathbf{B}$	$\mathbf{B}^\top \otimes \mathbf{I}$	multiply by a Kronecker matrix formed from $\mathbf{B}$
$\mathbf{C} \odot f_1$	$\operatorname{diag}(\operatorname{vec}(\mathbf{C}))$	multiply by a diagonal formed from $\mathbf{C}$
$f_1 / \mathbf{C}$	$\operatorname{diag}(\operatorname{vec}(\mathbf{C}))^{-1}$	multiply by the reciprocal diagonal

2.1.2 Example

We continue the example of §1.2.3, which canonicalized problem (1.5) into the conic-form trees of Figure 1.3. We now compute the problem data of these trees. The variables are first stacked into $\mathbf{z} = (t_1, t_2, \mathbf{x})$. Each variable then occupies a contiguous block of columns of $\mathbf{A}_f$, starting at its column offset within $\mathbf{z}$. The columns of the variables that do not appear in the expression f are zero, because f does not depend on their entries. Figure 2.2 gives an example of column offsets on another set of variables.

Matrix stuffing then runs bottom-up over the objective and the three constraint trees, as shown in Figure 2.3. Each leaf of $\mathbf{x}$ contributes an identity block of size n at the column offset of $\mathbf{x}$, and the constant leaf $\mathbf{g}$ contributes its value to the last column. The interior nodes

$$\mathbf{z}^\top = \begin{array}{cccccccc} & \mathbf{u} & & & \mathbf{v} & & & \mathbf{w} \\ & 0 & & & 2 & & & 5 \\ [& u_1 & u_2 & v_1 & v_2 & v_3 & w_1 & w_2] \end{array}$$

Figure 2.2 The column offsets of a stack of three variables $\mathbf{z} = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ with $\mathbf{u} \in \mathbb{R}^2$, $\mathbf{v} \in \mathbb{R}^3$, and $\mathbf{w} \in \mathbb{R}^2$. Each offset is the total dimension of the preceding variables: $\mathbf{u}$ starts at column 0, $\mathbf{v}$ at 2, and $\mathbf{w}$ at 5. The entries of $\mathbf{z}$ index the columns of every augmented matrix $[\mathbf{A}_f \ \mathbf{b}_f]$.

apply their operators to the matrices of their children. The product node $\mathbf{Fx}$, for example, multiplies the identity block of its argument by $\mathbf{F}$. Finally, the root stacks the augmented matrices of its children at their constraint row offsets into $[\mathbf{A}\ \mathbf{b}]$.

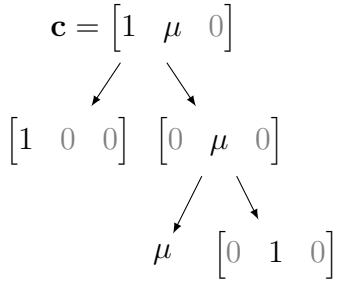
The augmented matrices of (2.2) can be stored in two ways, shown in Figure 2.4. In the global layout, drawn in Figure 2.3, each node stores one matrix whose columns span the whole stacked vector $\mathbf{z}$, with the column offsets applied from the start. In the local layout, which `CVXPY` uses [64], each node keeps one coefficient block per variable in its subtree, and one for the constant data. The offsets are applied only after the pass, when each block is placed at its variable's columns and the blocks are concatenated into the full matrix $[\mathbf{A}\ \mathbf{b}]$. The local layout stores only the columns of the variables present in a subtree. The tradeoff is that each interior node applies its operator once per stored block, rather than once per argument.

2.1.3 Parameter caching

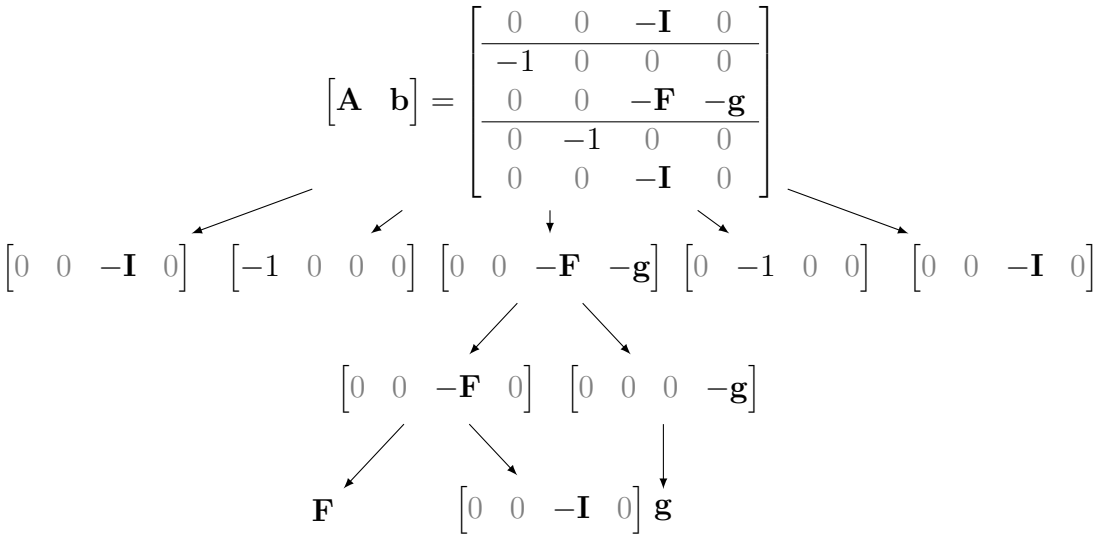
In a problem family the structure is fixed and only the numeric parameter values change between solves. Under additional rules on how the parameters enter the expressions, known as DPP, the problem data is affine in the parameters [26]. The rules classify each parameter as affine and forbid products of two parametrized expressions, so that the dependence can be represented by the maps of (2.5). To keep track of the parameter dependence, `CVXPY` adds a third dimension to the problem data, indexed by the parameter entries. Every coefficient matrix throughout the bottom-up pass becomes a tensor. At the root nodes we thus get the cached parameter maps

$$\mathbf{c} = \mathbf{Q}\,\tilde{\boldsymbol{\theta}}, \qquad \operatorname{vec}([\mathbf{A}\ \mathbf{b}]) = \mathbf{R}\,\tilde{\boldsymbol{\theta}}, \qquad \tilde{\boldsymbol{\theta}} = \begin{bmatrix} \boldsymbol{\theta} \\ 1 \end{bmatrix}, \tag{2.5}$$

where $\mathbf{Q} \in \mathbb{R}^{n \times (p+1)}$ and $\mathbf{R} \in \mathbb{R}^{m(n+1) \times (p+1)}$. The objective map $\mathbf{Q}$ is shown in Figure 2.5 for (1.5). After a parameter change, a re-solve costs one sparse matrix-vector product per map.



(a) Matrix stuffing of the objective



(b) Matrix stuffing tree of the constraints

Figure 2.3 Matrix stuffing of (1.5) by a bottom-up pass over the conic-form expression trees of Figure 1.3: each node's matrix is obtained by applying a linear operation to the matrices of its children. The coefficients appear negated because (1.2) writes the constraints as $\mathbf{b} - \mathbf{A}\mathbf{x} \in \mathcal{K}$. The horizontal rules in $[\mathbf{A}\ \mathbf{b}]$ separate the rows of the three constraints.

$$\left\{ \begin{array}{cc} \mathbf{x} & 1 \\ \mathbf{x} \mapsto [-\mathbf{F}], & -1 \mapsto [-\mathbf{g}] \end{array} \right\}$$

(a) Local layout.

$$\begin{array}{cccc} t_1 & t_2 & \mathbf{x} & 1 \\ [0 & 0 & -\mathbf{F} & -\mathbf{g}] \end{array}$$

(b) Global layout.

Figure 2.4 What each layout stores during the pass, at the node $-(\mathbf{F}\mathbf{x} - \mathbf{g})$ of (1.5): a dictionary of narrow per-variable blocks, or a single matrix spanning all columns.

$$\mathbf{Q} = \begin{array}{c} \\ t_1 \\ t_2 \\ \mathbf{x} \end{array} \begin{array}{c} \mathbf{g}^\top \quad \mu \quad 1 \\ \begin{bmatrix} 0 \cdots 0 & 0 & 1 \\ 0 \cdots 0 & 1 & 0 \\ 0 \cdots 0 & 0 & 0 \end{bmatrix} \end{array} \qquad \mathbf{c} = \mathbf{Q}\,\tilde{\boldsymbol{\theta}} = \begin{bmatrix} 1 \\ \mu \\ 0 \end{bmatrix}$$

Figure 2.5 The DPP path on the objective data of (1.5): the parameter map $\mathbf{Q}$ is built and cached once, and each re-solve recovers $\mathbf{c}$ with the single sparse product $\mathbf{Q}\tilde{\boldsymbol{\theta}}$.

Problems that do not conform to the DPP grammar, or that opt out of it with the option `ignore_dpp`, take a different approach to matrix stuffing. The reduction `EvalParams` replaces each parameter leaf by a constant holding its current value, and the resulting parameter-free tree goes through the pass as if the problem had no parameters, as shown in Figure 2.6. The result is the same objective data $\mathbf{c}$ as in Figure 2.5, now obtained by repeating the whole matrix stuffing instead of applying the cached parameter map $\mathbf{Q}\tilde{\boldsymbol{\theta}}$. The `ignore_dpp` path is more general, as it supports arbitrary functions of the parameters.

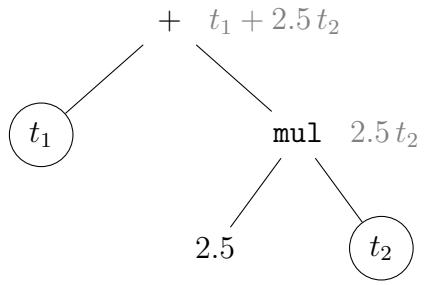


Figure 2.6 The `ignore_dpp` path on the objective of (1.5): the parameter leaf μ is replaced by a constant holding its current value 2.5, and the pass is repeated on the parameter-free tree on every change of the parameters.

`CVXPY` implements the pass in three canonicalization backends that differ in how they represent the parameter axis [65]:

- the C++ backend `cvxcore` keeps one sparse matrix per parameter slice and loops over the slices at every operator.
- the `SCIPY` backend stacks the slices vertically into a single strided two-dimensional sparse matrix. The backend is named after the `scipy.sparse` library that the matrix is stored in (we write `SCIPY` for the backend and `scipy.sparse` for the library).
- the `COO` backend does not form the tensor explicitly. It is named after the COO format

and extends its (row, column, value) triples with a parameter index (we write `COO` for the backend and COO for the format).

When no backend is requested, `CVXPY` uses `cvxcore`, and substitutes `COO` for a DPP problem with at least 1000 parameter entries in total. When canonicalization remains too slow, `CVXPYgen` can compile a DPP problem ahead of time [66]. It generates C code specific to the problem family, with the entries of the cached maps (2.5) written into the source as constants, so a re-solve applies the parameter maps in compiled code.

The pass of this section produces the affine data $\mathbf{c}$ and $[\mathbf{A}\ \mathbf{b}]$. For a quadratic objective, `CVXPY` replaces the affine argument of each quadratic atom by a new variable, adding the corresponding equality constraint. The objective becomes a sum of quadratic forms with constant matrices. A second matrix stuffing pass then assembles these matrices into the diagonal blocks of the matrix $\mathbf{P}$ of (1.2) [64].

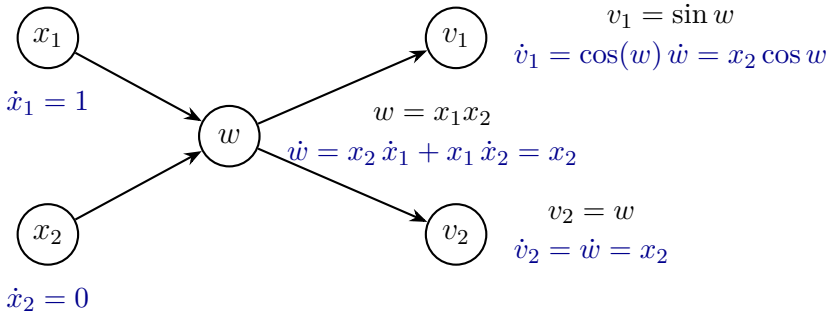
All three backends share two limitations. Firstly, they delegate their matrix operations to a general-purpose sparse library, `Eigen` [60] in `cvxcore` and `scipy.sparse` [58] in the Python backends. The storage formats and kernels are fixed by the library, so structure known at canonicalization time cannot be used. In `cvxcore`, the views of Table 2.1 also materialize explicit sparse operators. For example, a reshape builds a sparse identity matrix and multiplies the coefficients by it. Secondly, the canonicalization backends store dense constant matrices as sparse matrices. When dense blocks appear in the problem data, they are both stored in an inconvenient format and computed without optimized BLAS kernels. Our approach in Chapter 3 addresses these limitations.

2.2 Sparse derivatives from detected structure

This section reviews methods that compute the full sparse Jacobian of a vector-valued function, given as an arbitrary program in a general-purpose language. No modelling language declares the structure of the program, so the sparsity of the derivatives must be detected [67]. Modern surveys of AD address dense derivatives, or the gradient of a scalar function, which reverse mode computes in a single sweep over the computational graph [41, 68]. In those settings, AD refers to the evaluation of matrix-vector products with the Jacobian or the Hessian, and the explicit matrix is never formed. Factorizations, direct linear solves, and many optimization solvers instead require the explicit matrix [67]. The methods of this section recover the full sparse matrix from multiple matrix-vector products [47]. To reduce the number of products, graph colouring groups the vectors that can be propagated together, so that one product recovers several columns at once.

2.2.1 Matrix-vector products and accumulation

AD evaluates derivatives by sweeps of the computational graph. A sweep visits every node once and applies the chain rule at each, at about the cost of one function evaluation [47]. Each sweep computes one matrix-vector product with the derivative. Forward mode starts from a seed vector $\mathbf{v}$ at the inputs and propagates tangents, and the Jacobian-vector product (JVP) $\mathbf{Jv}$ appears at the outputs. Reverse mode starts from a seed $\mathbf{w}$ at the outputs and propagates adjoints, and the vector-Jacobian product (VJP) $\mathbf{J}^\top\mathbf{w}$ appears at the inputs, as shown in Figures 2.7 and 2.8. For the Hessian, the analogous Hessian-vector product $\mathbf{Hv}$ comes from one forward sweep over one reverse sweep.



$$\begin{bmatrix}\dot{v}_1\\ \dot{v}_2\end{bmatrix} = \begin{bmatrix}x_2\cos w\\ x_2\end{bmatrix} = \mathbf{J}\mathbf{e}_1, \qquad \mathbf{J} = \begin{bmatrix}x_2\cos w & x_1\cos w\\ x_2 & x_1\end{bmatrix}.$$

Figure 2.7 Forward sweep of the computational graph of $F(x_1, x_2) = (\sin(x_1x_2),\ x_1x_2)$, with the intermediate value $w = x_1x_2$. Alongside its value, each node carries one tangent $\dot{v}$ (in blue), updated by the chain rule following the order of evaluation. The seed $(\dot{x}_1, \dot{x}_2) = (1, 0)$ delivers the first column of $\mathbf{J}$ at the outputs.

Recovering the full matrix takes a sequence of sweeps, one per seed vector. The n unit seed vectors recover $\mathbf{J}$ column by column in forward mode, and the m unit seed vectors recover it row by row in reverse mode [47]. The other approach is accumulation [69], which combines the local derivatives of the graph directly into $\mathbf{J}$ by eliminating one intermediate vertex at a time, as shown in Figure 2.9. Finding the optimal elimination order is NP-complete [47, 70]. Batching the n unit seeds into one forward sweep gives each node a whole tangent, the derivative of that node with respect to all the inputs. Eliminating the vertices in the order of evaluation stores the same derivative at each node, so the batched sweep and forward-order elimination perform the same computation.

In practice, AD tools such as `ADOL-C` [71] and `CppAD` [72] record the scalar graph as a runtime trace, one record per elementary operation [47, 73]. A single matrix-vector product unrolls

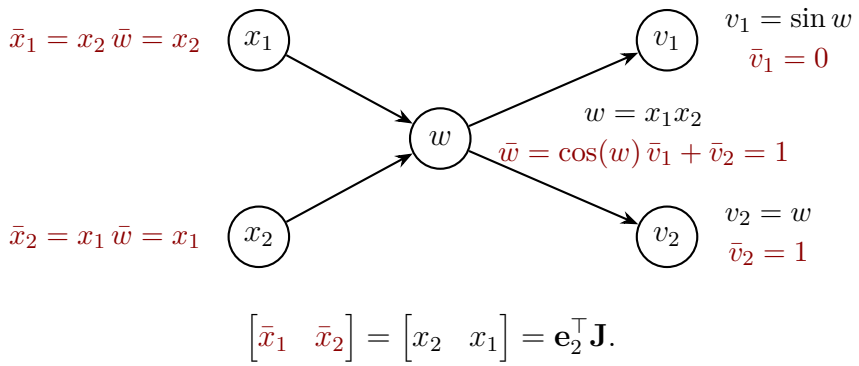
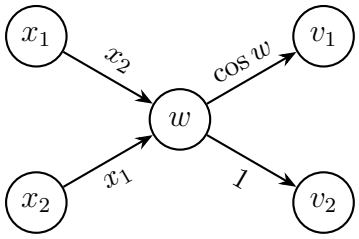
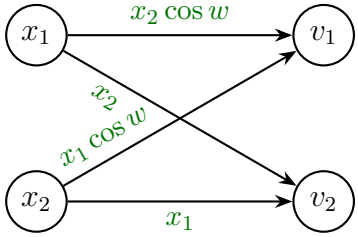


Figure 2.8 Reverse sweep of the computational graph of Figure 2.7. Each node carries one adjoint $\bar{v}$ (in red), updated by the chain rule against the order of evaluation. The seed $(\bar{v}_1, \bar{v}_2) = (0, 1)$ delivers the second row of $\mathbf{J}$ at the inputs.



(a) Each edge carries a local derivative.



(b) After eliminating w, the edges are the entries of $\mathbf{J}$.

Figure 2.9 Accumulation of the Jacobian by vertex elimination, on the graph of Figure 2.7. Eliminating the intermediate vertex w connects each of its predecessors to each of its successors and multiplies the local derivatives along the path (in green). Parallel edges, when they arise, are summed. Once every intermediate vertex is eliminated, the graph is bipartite and its edges are the entries of $\mathbf{J}$.

into one scalar multiplication for every entry of the matrix, together with the additions that sum them, so the trace of a realistic problem can sometimes reach millions of nodes. On such a trace, every intermediate vertex would store its derivative with respect to all the inputs, so accumulation is rarely used in practice. When $\mathbf{J}$ is sparse, most of each sweep is then wasted on zeros, and the cost of recovering the matrix scales with its dimension rather than with its number of nonzeros.

2.2.2 Compression by graph colouring

Several columns of a sparse Jacobian can share one seed [28]. Two columns with no nonzero in a common row are structurally orthogonal, and one sweep recovers both, each entry landing

in a distinct row of the product. Finding the fewest such groups was later formalized as a colouring of the column intersection graph, which has one vertex per column and an edge between two columns that share a row [29, 74]. Figure 2.10 colours the graph of a 6×8 Jacobian and recovers the matrix in three sweeps instead of eight. The approach extends to the Hessian, which compresses further because its symmetry allows dedicated star and acyclic colourings [75, 76].

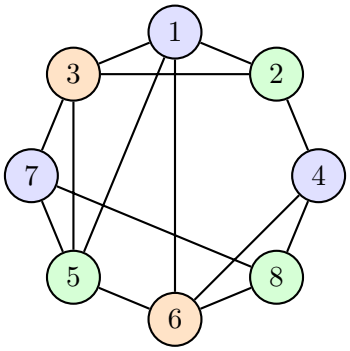
The sparse Jacobian is obtained from the products in three stages: detect the sparsity pattern, compress it by colouring, and propagate the compressed seeds of Figure 2.10 through the graph to recover the nonzero entries [67]. Detection propagates sparsity patterns instead of values through the same graph, replacing each arithmetic operation by a union of index sets [47]. The pattern and its colouring are computed once and reused, so their cost is spread across repeated evaluations of the same function.

2.2.3 `CasADi`

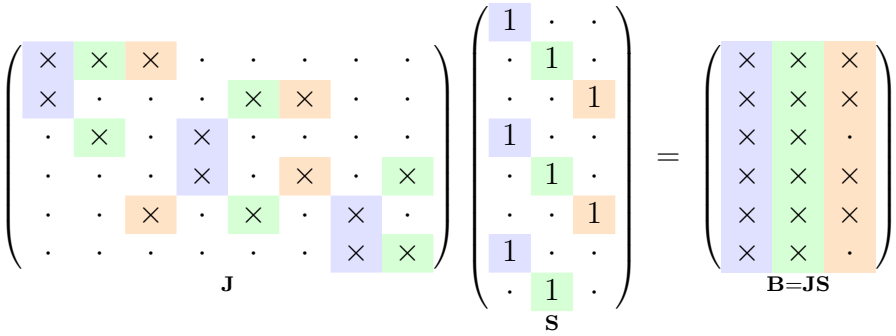
`CasADi` [12] is an open-source framework for nonlinear optimization and optimal control built around AD. It is widely used in academia and industry, notably for model predictive control, trajectory optimization, robotics, and energy systems [77]. Its core is written in C++ and is used from Python, MATLAB, or Octave. It provides the building blocks of gradient-based optimization: symbolic expressions, derivatives, integrators, and interfaces to NLP and QP solvers. The gradients, Jacobians, and Hessians these solvers require are obtained by differentiating the model.

`CasADi` supports two types of expressions, and both return the matrix derivatives in a sparse format, which is expected by most solvers [77]. An `SX` graph is scalar, with exact node-by-node sparsity, but its size grows with the problem data. An `MX` graph is vectorized. Its nodes are whole matrix operations, so the graph stays small even when the model is large. Either graph can be exported as self-contained C code and compiled ahead of time for embedded applications.

`CasADi` computes its derivatives by the detection, colouring, and batched sweeps of this section, and caches the pattern, the colouring, and the compiled derivative function across evaluations. Listing 2.1 rebuilds the conic-form trees of Figure 1.3, obtained after canonicalization, as `MX` expressions. Differentiating them recovers the problem data that matrix stuffing assembles in §2.1.2.



(a) The column intersection graph of **J**, with one vertex per column and an edge between two columns that share a row, coloured with three colours.



(b) The seed matrix **S** sums the unit vectors of each colour class, and the compressed product $\mathbf{B} = \mathbf{JS}$ recovers every nonzero of **J**, each in a distinct row of its compressed column.

Figure 2.10 Graph colouring compresses the seeds. Colouring the column intersection graph groups the columns of the sparse Jacobian **J** into three structurally orthogonal classes, and the matrix is recovered in three sweeps instead of eight.

```
import casadi as cs
t1, t2 = cs.MX.sym("t1"), cs.MX.sym("t2")
x = cs.MX.sym("x", n)
g, mu = cs.MX.sym("g", m), cs.MX.sym("mu")
z = cs.vertcat(t1, t2, x)                # stacked variables
expr = cs.vertcat(x, t1, F @ x - g, t2, x) # stacked affine expressions
A = -cs.jacobian(expr, z)                # the matrix A
b = cs.substitute(expr, z, 0)            # the vector b
c = cs.gradient(t1 + mu * t2, z)         # the objective vector c
```

Listing 2.1 The conic form of the running example in `CasADi`.

The Jacobian with respect to the stacked variable $\mathbf{z}$ recovers the matrix $\mathbf{A}$ of Figure 2.3 entry by entry. Sparse differentiation is the equivalent of the matrix stuffing of §2.1, with the coefficients obtained by these sweeps rather than by a bottom-up pass over the tree. The parameters remain symbols, so $\mathbf{b}$ and $\mathbf{c}$ are expressions of $\mathbf{g}$ and μ, re-evaluated for each parameter value. This is how the `qpsol` interface builds its problem data internally, for problems with a quadratic objective and affine constraints. A model written at the same high level as in `CVXPY`, with the norms of (1.5) kept in the expression graph, is solved through `nlpsol` instead. The same sweeps assemble the gradient, constraint Jacobian, and Hessian of the Lagrangian that the NLP solver queries at each iteration.

2.3 Coefficient extraction from declared structure

In contrast to §2.2, the methods of this section require the user to declare the structure of the problem. In `AMPL` and `JuMP`, the user writes each objective and constraint as a sum of terms, and the coefficient of every linear and quadratic term is stored as it is typed. Neither language has an atom library or a canonicalization step, so a nonlinear expression such as a norm is left to the solver (`AMPL`) or written in conic form by the user (`JuMP`). In place of differentiation, `AMPL` writes the stored coefficients to a `.nl` file that the solver parses, and `JuMP` keeps them as typed in-memory term lists that are copied to the solver.

2.3.1 `AMPL`

`AMPL` [5] is a commercial algebraic modelling language for large-scale mathematical programming, developed at Bell Laboratories in the 1980s. It is one of the most widely used modelling languages in academia and industry, applied to production planning, scheduling, logistics, and energy systems [78]. A model is written declaratively, close to its mathematical statement, from sets, indexed parameters, indexed variables, and algebraic expressions over them. Data is supplied separately, so one model can be used for many problem instances. The name `AMPL` denotes both the modelling language and the commercial system that processes it. Models are written in the language directly or built through its Python interface [79], and reach a large collection of commercial and open-source solvers through a single instance format.

The `AMPL` system splits the work between two programs, which its documentation calls the translator and the driver. The translator runs as its own process. It parses the model and the data, expands every indexed declaration into individual variables and constraints with numeric coefficients, and writes the result to a `.nl` file [30]. The driver is a separate

executable that reads the `.nl` file and calls the solver, and each solver is reached through its own driver. A driver is built on one of two libraries. The AMPL Solver Library (ASL) is the original, a C library that parses the `.nl` format and evaluates the expressions it holds [31]. `MP` is the newer one, which `AMPL` supplies for solvers that need no derivatives [80]. This division between a translator that writes an instance and a driver that reads it still defines how most solvers receive nonlinear problems.

The `.nl` file is an instance format rather than a model format. Sets and indexing are resolved by the translator, and what remains is one flat record per objective and constraint. An indexed declaration such as `var x {1..n}` becomes n scalar variables, and an indexed constraint becomes one record per index, so the file has no notion of a vector. The linear part of each record is an explicit list of (variable, coefficient) pairs, stored alongside the bounds and the nonzero counts of the instance. Only the nonlinear parts are kept as expression graphs, written in prefix notation [30].

The language has no norm atom and no canonicalization step, so the norms of (1.5) are entered as square roots of sums of squares. What the solver receives depends on its driver. An `MP`-based driver rewrites the two norms into second-order cones for a conic solver, and an ASL-based driver passes on a general nonlinear program. Listing 2.2 states the model.

```
param m; param n;
param F {1..m, 1..n};
param g {1..m};
param mu >= 0;
var x {1..n} >= 0;
minimize obj:
    sqrt(sum {i in 1..m} (sum {j in 1..n} F[i,j]*x[j] - g[i])^2)
  + mu * sqrt(sum {j in 1..n} x[j]^2);
```

Listing 2.2 The running example (1.5) in `AMPL`.

For an instance with $m = 3$, $n = 2$, and $\mu = 0.1$, the translator writes the `.nl` file of Listing 2.3. The comments are added by hand to make the records readable and are not written by the translator.

```
g3 1 1 0    # text format; 9 more header lines follow
 2 0 1 0 0  # 2 vars, 0 constraints, 1 objective, 0 ranges, 0 eqns
...
O0 0        # objective 0, an expression graph in prefix notation
o0          #   add
o39         #     sqrt
o54         #       sum of 3 terms, one per row of F
...         #         affine subtrees F[i,1]*x[1] + F[i,2]*x[2] - g[i]
o2          #   multiply
n0.1        #     the value of mu
...         #     sqrt subtree of the second norm
b           # variable bounds: x[1] >= 0, x[2] >= 0
2 0
2 0
```

Listing 2.3 An abbreviated `.nl` file for an instance of the running example.

The whole objective is written as one expression graph, and the value of μ is folded into it as the constant `n0.1`, so a parameter change requires a new file. For a linear or quadratic program, no graph remains and the file contains the complete problem data as coefficient lists.

Inside the driver, the bounds and the linear coefficient lists are read directly from the file, at a cost linear in the number of nonzeros, and only the nonlinear parts are rebuilt as expression graphs. What happens to those graphs depends on the library. `MP` matches the second-order and exponential cones of §1.2.1 in the rebuilt graphs. A conic solver such as `Gurobi` or `MOSEK` then receives cone constraints rather than the square roots the model states. ASL instead differentiates the graphs by reverse-mode AD [81], with sparse Hessians assembled from partial separability [82].

Any system that writes `.nl` reaches every solver that reads it. The format has grown since, with extensions such as logical constraints, and `AMPL` omits the added header fields when a model does not use them, so older drivers still read the files it writes [30]. An instance is an ordinary file, so it can be archived, shipped to reproduce a bug, and rerun on another machine. However, serializing the instance to disk can be slow, and every solve repeats it, rebuilding any expression graphs from scratch.

2.3.2 JuMP

JuMP [11, 83] is an open-source algebraic modelling language embedded in Julia, developed at MIT and first released in 2013. It is the most widely used modelling language in Julia, applied in academia and industry to power systems, portfolio optimization, scheduling, and routing [83]. A model is written with the macros `@variable`, `@constraint`, and `@objective`. The abstraction layer between models and solvers is MOI [32], and through it the same model can be passed to any of more than fifty supported solvers [84].

The macros of JuMP rewrite each expression as it is entered. `@objective` and `@constraint` build affine and quadratic expression objects term by term, so the model stores coefficient lists instead of an expression tree. These lists are typed by MOI. A `VectorAffineFunction` is a list of (row, coefficient, variable) terms and a constant vector. A `ScalarQuadraticFunction` adds a list of (coefficient, variable, variable) triples. A constraint states that one such function lies in a set. For a nonnegative orthant or a second-order cone, this is exactly the functional conic form of §1.2.1. MOI also supports other sets, such as integrality, special ordered sets, and indicator constraints [85].

The user declares the epigraph variables and cone memberships of (1.6) directly, as in Listing 2.4.

```
using JuMP
model = Model()
@variable(model, x[1:n] >= 0)
@variable(model, t1); @variable(model, t2)
@constraint(model, [t1; F * x - g] in SecondOrderCone())
@constraint(model, [t2; x] in SecondOrderCone())
@objective(model, Min, t1 + mu * t2)
```

Listing 2.4 The running example (1.5) in JuMP.

The first cone constraint is stored as shown in Listing 2.5.

```
VectorAffineFunction([(1, 1.0, t1), (2, F[1,1], x[1]), ...],
                     [0.0, -g[1], ..., -g[m]]) in  SecondOrderCone(m + 1)
```

Listing 2.5 The first cone constraint as stored by MOI.

The expression itself is not retained after the macro returns.

When the model is passed to a solver, MOI copies each constraint into the solver's own representation with a `copy_to` function. Solvers that want matrix data commonly implement that copy through the `MatrixOfConstraints` utility, which concatenates the stored term lists

into one sparse matrix in the solver's format [85], with no expression graph to visit and no differentiation. Only the nonlinear path of `JuMP` retains expression graphs, and differentiates them by sparse AD.

When a solver does not support a constraint or objective type natively, MOI reformulates it through its bridge system. A bridge is a structural rewrite from one function-set pair to another, for instance from a rotated to a standard second-order cone. For a conic solver, a bridge rewrites a quadratic objective into a second-order cone constraint with an epigraph variable. MOI selects and composes bridges automatically [32]. Bridging manipulates stored coefficients and adds auxiliary variables and constraints. A parameter is a variable in the `Parameter` set. A parameter multiplied by a variable is stored as a quadratic term, so keeping such a model affine requires `ParametricOptInterface.jl`, an extension that substitutes the parameter values before the solve [84].

2.4 Research gap

The four modelling languages reviewed in this chapter obtain the problem data in one of two ways, by differentiating an expression graph or by reading the coefficients the user declares. `CasADi` is built for nonlinear optimization. Its conic interface, `qpsol`, supports QPs with affine constraints and obtains their coefficients by differentiating the expression graph, with detected sparsity and seeds compressed by colouring. `CVXPY` composes the linear operators of the conic-form trees in one bottom-up pass, which is equivalent to differentiating them. Both keep the model as an expression graph between solves. `AMPL` and `JuMP` take the coefficients as the user declares them, written to the `.nl` file by the translator or stored as term lists by the macros. In both, only nonlinear expressions are kept.

The four modelling languages also differ in how they handle parameter changes. `CasADi` keeps the parameters as symbols and splits the work in two phases. The derivative function is built once, and a parameter change requires only its evaluation, one sweep per colour. `CVXPY` caches the parameter-to-data tensor of §2.1.3 and re-evaluates with one tensor-vector product, but only for problems conforming to the DPP ruleset. For every other problem nothing is cached and the whole pass is repeated. `AMPL` folds the values into the instance, so a change requires the translator to write a new `.nl` file and the driver to read it. `JuMP` modifies the term lists in place.

None of the four combines the two-phase split with direct computation of the problem data. Chapter 3 presents a new method that does. It splits the work in two phases, just like `CasADi`, but instead of matrix-vector products it computes the entire coefficient matrix directly, in one

bottom-up pass over the conic-form trees. Compared with the `CVXPY` backends, our proposed method also uses the dense structure of the problem, with dense storage and computation instead of general-purpose sparse matrix libraries. Table 2.2 summarizes these differences.

Table 2.2 How the four modelling languages and `SparseDiffEngine` obtain and reuse the coefficients.

System	**Expression storage**	**Coefficients obtained by**	**Reused across the family**	**Repeated at every solve**
`AMPL`	scalar	instance generation	nothing, values folded into the instance	`.nl` write and read
`JuMP`	scalar	term lists stored by macros	term lists, modified in place	`copy_to` into the solver
`CasADi`	scalar (`SX`) or matrix (`MX`)	differentiation of an expression graph	pattern, colouring, derivative function	one sweep per colour
`CVXPY` backends	vectorized	composition of linear operators	parameter-to-data tensor, DPP problems only	one tensor-vector product, or the whole pass
`SparseDiffEngine` (Chapter 3)	vectorized	structured direct differentiation	problem structure, by the symbolic pass	numeric pass

CHAPTER 3 A LIGHTWEIGHT CANONICALIZATION ENGINE

This chapter presents `SparseDiffEngine`, our C library for assembling the constraint and objective matrices of a canonicalized problem. The engine computes the same derivatives as `CVXPY`'s canonicalization backends. Both apply the chain rule of matrix calculus to trees of vectorized expressions. Our contributions lie in the implementation. Firstly, we split the computation into a symbolic pass, which computes and caches the structure of every intermediate matrix, and a numeric pass, which fills in values. This separation is particularly useful for compiling problems involving parameters, where the structure is fixed but the numerical values can change. Secondly, the data of most optimization problems is not only sparse but also structured, with dense blocks and diagonal patterns. We introduce two new matrix classes for this purpose, `permuted-dense` and `stacked-permuted-dense`, which let us use dense storage and BLAS subroutines. Neither the two-phase approach nor the two matrix classes exists in `CVXPY`'s backends. Together, they address the research gap we identify in §2.4.

Other important design decisions differ from `CVXPY`'s backends for simplicity rather than performance. Every node carries one Jacobian taken with respect to all of the variables of the problem, rather than a dictionary of local Jacobians keyed by variable. As a consequence, the chain rule is a single matrix product, and the constraint matrix is built by stacking the Jacobians of the constraints. In addition, quadratic functions have a dedicated Hessian pass, instead of a variable substitution followed by a second matrix stuffing pass.

The rest of the chapter is organized as follows. We begin with a brief overview of the engine, describing how expressions are represented and how the code is organized. We then present the three matrix representations used for the derivative computations. Next, we describe the algorithms, or kernels, which operate on and combine these matrix representations. Finally, we describe the problem interface of the engine, and how `CVXPY` obtains and updates the problem data through it. We conclude with a summary.

3.1 Overview of the engine

`SparseDiffEngine` is a small C library, around seventeen thousand lines of code and a similar number of lines for the test suite. The expression node is defined in `expr.h`. Atoms that carry extra data, such as the exponent of a power, define a larger structure in `subexpr.h`. We implement each atom in its own file in the `atoms/` directory. There are around forty-five

of them, divided into six subdirectories based on their mathematical properties. The matrix data structures and the computational kernels that operate on them are in `utils/`, and `problem.c` collects the objective and constraints of a problem and evaluates the values and derivatives that a solver asks for.

3.1.1 Atom classes

Atoms in the same subdirectory compute their derivatives in the same way. Table 3.1 lists the classes and how each computes its Jacobian. Canonicalization leaves the constraints affine and the objective affine or quadratic. The quadratic atoms of a DCP problem, such as a squared power, a quadratic over a linear term, or a Huber penalty, are all rewritten as quadratic forms by the canonicalization frontend of §1.2.2. Therefore, only affine atoms and quadratic forms reach the engine from a canonicalized DCP problem. The rest of the chapter describes the affine atoms. The quadratic form contributes the objective matrix $\mathbf{P}$ through the Hessian, described in §3.4. The remaining classes are used by the NLP interface of `CVXPY` [34]. Each atom implements three operations. The forward pass evaluates the atom at a point. The symbolic pass computes the sparsity patterns of its derivatives and allocates memory for them. The numeric pass computes the numerical values inside the fixed sparsity patterns. Each atom also has a constructor, the C function that allocates and initializes the corresponding expression node. The constructor stores pointers to these three operations in the node.

Table 3.1 The atom classes and how each computes its Jacobian.

Class	Examples	Jacobian computation
affine	`sum`, `index`, `left_matmul`	fixed, by index maps and copies
elementwise, full domain	`exp`, `power`, `logistic`	diagonal scaling of the child Jacobian
elementwise, restricted domain	`log`, `entr`, `atanh`	diagonal at the variable's columns
bivariate, full domain	`multiply`, `matmul`	two-term chain rule
bivariate, restricted domain	`quad_over_lin`, `rel_entr`	two-term chain rule
other	`quad_form`, `prod`	specific to each atom

3.1.2 Expression node

An expression node stores its dimensions, pointers to at most two children, a buffer for its value, and its derivative matrices. Every derivative matrix has `n_vars` columns, the total number of entries of all the variables of the problem. Listing 3.1 shows a simplified version of the node.

```
typedef struct expr
{
    int d1, d2, size, n_vars, refcount, var_id;
    struct expr *left;
    struct expr *right;

    double *value;   /* forward value             */
    matrix *jacobian; /* Jacobian w.r.t. the variables */

    forward_fn      forward;
    jacobian_init_fn jacobian_init_impl; /* symbolic pass */
    eval_jacobian_fn eval_jacobian;    /* numeric pass */
} expr;
```

Listing 3.1 A simplified version of the expression node in `expr.h`.

Trees are built bottom up, with one constructor per atom. A `CVXPY` expression is converted in one pass over the tree, which calls the constructor of the corresponding atom at each node, as described in §3.4. Expressions that appear multiple times can be shared, in which case the tree becomes a directed acyclic graph. Nodes are therefore reference counted, and the symbolic pass returns early if a node was already initialized.

The engine has only two kinds of leaf node, variables and parameters, where `CVXPY` also has constants. A variable occupies a contiguous slice of the stacked variable vector, starting at the column offset `var_id`. Its Jacobian is an identity block at the columns of the variable, written once during the symbolic pass. The Jacobian of a parameter is empty, because parameters are fixed during a solve. The engine treats a constant as a parameter whose value cannot be updated between solves.

3.2 Matrix representations

We store the Jacobian of a node in one of three representations, a generic sparse matrix, a permuted dense matrix, or a stacked permuted dense matrix. The generic sparse matrix is

the default. The other two are created by matrix multiplication and the quadratic form, the operations whose data is dense. Each implements the same operations, so an atom picks one when it allocates its Jacobian during the symbolic pass.

3.2.1 Sparse matrix

The generic sparse representation stores a Jacobian in the CSR format of §1.4.2. A Jacobian has one row for each entry of the value of its node, and most operations, such as indexing, broadcasting, and stacking, operate on those rows. Each row is stored contiguously and reached in constant time. For this reason we chose the CSR format over the CSC format. In `SparseDiffEngine` the column indices within each row are kept sorted. The CSR format does not necessarily require this order, and libraries differ on it. For example, of the two sparse libraries behind `CVXPY`'s backends, `Eigen` keeps its indices sorted [60], while `scipy.sparse` does not always enforce it [58]. The merges and intersections of the symbolic pass depend on this sorted order guarantee.

The symbolic pass allocates the row pointer and column index arrays, and the numeric pass writes only the value array. §3.3.2 describes what this split between the sparsity pattern and the values saves across solves.

3.2.2 Permuted dense matrix

The Jacobians that appear during canonicalization do not always have scattered, isolated nonzeros. Multiplying an expression by a data matrix, for example, places a dense block at a subset of the rows and columns of the Jacobian. Consider the expression $\mathbf{Bx}$ with dense data. Its Jacobian holds the data at the columns of the variable and is zero everywhere else, as visualized in Figure 3.1. Those columns are usually a small part of the Jacobian. The generic sparse representation would instead need to store a column index for each entry of the block.

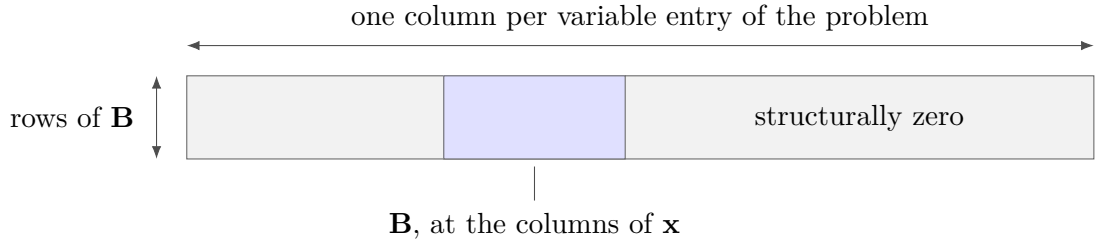


Figure 3.1 The Jacobian of $\mathbf{Bx}$ with respect to the full variable vector, for dense data $\mathbf{B}$. Its nonzeros are exactly the entries of $\mathbf{B}$.

The dense block can instead be stored together with the indices of the rows and columns it occupies, one index per row and column rather than one per entry, and the global matrix is never formed. We define a permuted dense matrix as the triple $(\mathbf{X}, \mathbf{r}, \mathbf{c})$, with $\mathbf{X} \in \mathbb{R}^{m_0 \times n_0}$ dense and $\mathbf{r}$ and $\mathbf{c}$ strictly increasing index vectors, representing

$$\mathbf{A} = \mathbf{S}_r^\top \mathbf{X} \mathbf{S}_c \in \mathbb{R}^{m \times n}, \qquad A_{r_i, c_j} = X_{ij}, \tag{3.1}$$

where the rows of $\mathbf{S}_r \in \{0,1\}^{m_0 \times m}$ and $\mathbf{S}_c \in \{0,1\}^{n_0 \times n}$ are the rows of the identity indexed by $\mathbf{r}$ and $\mathbf{c}$, as shown in Figure 3.2. Every other entry of $\mathbf{A}$ is structurally zero and never stored. The entries of $\mathbf{c}$ are columns of the global variable vector and the entries of $\mathbf{r}$ are rows of the Jacobian.

$$\mathbf{A} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{20} & b_{21} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{30} & b_{31} & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \qquad \mathbf{X} = \begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \\ b_{20} & b_{21} \\ b_{30} & b_{31} \end{pmatrix}$$

$$\mathbf{r} = (0, 1, 2, 3), \qquad \mathbf{c} = (4, 5).$$

$$\mathbf{A} = \underbrace{\begin{pmatrix} 1 & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & 1 \end{pmatrix}}_{\mathbf{S}_r^\top} \underbrace{\begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \\ b_{20} & b_{21} \\ b_{30} & b_{31} \end{pmatrix}}_{\mathbf{X}} \underbrace{\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \end{pmatrix}}_{\mathbf{S}_c}$$

Figure 3.2 The Jacobian of $\mathbf{Bx}$ as a permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{4 \times 2}$ and a variable $\mathbf{x} \in \mathbb{R}^2$ at columns $(4, 5)$ of a problem with ten variable entries. Only the dense block $\mathbf{X}$ is stored, and the selectors are the rows of the 4×4 and 10×10 identities indexed by $\mathbf{r}$ and $\mathbf{c}$.

The dense block is stored in row-major order and the index vectors as two integer arrays, so the storage does not depend on the global dimensions m and n. Structural atoms, which only rearrange entries, turn a permuted dense matrix into another permuted dense matrix by manipulating the row and column indices alone. The transpose of a permuted dense matrix swaps the two index vectors and transposes the dense block. Figure 3.3 indexes the first and last rows of the Jacobian of $\mathbf{Bx}$. The row vector of the result is $(0, 1)$ rather than $(0, 3)$, because it indexes the rows of the result rather than of the original block. The map back to the rows of the original block is kept internally, and the numeric pass copies the values through this map at every evaluation.

$$\mathbf{A}[(0,3),\,:] \;=\; \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{30} & b_{31} & \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

$$\text{represented by the triple} \quad \left(\begin{pmatrix} b_{00} & b_{01} \\ b_{30} & b_{31} \end{pmatrix},\; (0,1),\; (4,5) \right).$$

Figure 3.3 Row indexing of the $\mathbf{A}$ of Figure 3.2. The result is mostly zeros, but its nonzeros still form one smaller dense block.

A permuted dense matrix has nonzeros only inside its dense block. The Jacobian of $\mathbf{Bx} + \mathbf{y}$ contains the block of $\mathbf{B}$ at the columns of $\mathbf{x}$ and an identity at the columns of $\mathbf{y}$, which no single block covers. The sum is instead performed on CSR views of the two operands. A view is built during the symbolic pass, and its value array points to the dense block, so no values are copied. The result is stored as a generic sparse matrix, and so is everything computed from it.

3.2.3 Stacked permuted dense matrix

Multiplying a matrix-valued expression with n rows and p columns by data $\mathbf{B} \in \mathbb{R}^{m \times n}$ multiplies each of the p columns by $\mathbf{B}$. On the vectorized expression the multiplication acts as the block diagonal operator $\mathbf{I}_p \otimes \mathbf{B}$ of Table 2.1. In practice this operator is not formed, unlike in `CVXPY`'s backends. The atom stores a single copy of $\mathbf{B}$ and applies it to one column at a time. Its Jacobian contains one dense block for each of the p columns, at pairwise disjoint rows and in general at different columns, as in Figure 3.4. No single triple $(\mathbf{X}, \mathbf{r}, \mathbf{c})$ describes such a matrix. Merging the blocks into one larger block would store the zeros between them, and falling back to the generic sparse representation would lose the dense structure. We instead keep the blocks as they are, in a third representation.

We define a stacked permuted dense matrix as a vertical union of permuted dense blocks whose row vectors are pairwise disjoint, while the column vectors may overlap. Every operation treats the blocks one at a time, and the global matrix is never formed. The stacked matrix owns a single value buffer, and each block points into it.

3.3 Computational kernels

An atom does not operate on the representations of §3.2 directly. It instead calls a kernel, the implementation of one operation for one matrix representation. The ten operations of Table 3.2 cover the whole library, so every atom computes its derivatives through a small

$$\mathbf{M} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot \end{pmatrix}$$

$$\mathbf{X}_1 = \mathbf{X}_2 = \mathbf{B}, \qquad \mathbf{r}_1 = (0, 1), \quad \mathbf{c}_1 = (4, 5), \qquad \mathbf{r}_2 = (2, 3), \quad \mathbf{c}_2 = (6, 7).$$

Figure 3.4 The Jacobian of $\mathbf{BV}$ as a stacked permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{2\times 2}$ and a variable $\mathbf{V} \in \mathbb{R}^{2\times 2}$ at columns 4 to 7 of a problem with ten variable entries. There is one block per column of $\mathbf{V}$, and the 4×10 matrix $\mathbf{M}$ is never formed in practice.

set of shared routines. We implement each kernel as a pair of functions, one called by the symbolic pass and one by the numeric pass.

3.3.1 Kernel dispatch

The operations of Table 3.2 divide into unary operations, which operate on a single matrix, and binary operations, which combine two matrices. A call on a Jacobian must select the kernel that matches its representation, a step called dispatch. For a unary operation, each representation implements the operation and the call runs that implementation. The kernel of a binary operation may depend on the representations of both operands.

For example, on a generic sparse matrix the symbolic pass of the `index` kernel allocates a new CSR pattern and copies the column indices of the selected rows. On a permuted dense matrix it intersects the selected rows with the row index vector, exactly as in Figure 3.3. On a stacked matrix the permuted dense kernel is applied to each block, and the blocks left with no selected rows are dropped. In all three representations the numeric pass only copies the values of the rows that were kept.

Of the three binary operations, only the transposed products dispatch on both operands. Addition always operates in the CSR format. Both operands are read through their CSR views and the result is a generic sparse matrix, so no dispatch is needed. In the product $\mathbf{BA}$, the matrix $\mathbf{B}$ is data rather than a Jacobian, so only the representation of $\mathbf{A}$ selects the kernel. We select the kernel of the transposed products $\mathbf{B}^\top\mathbf{A}$ and $\mathbf{B}^\top\mathbf{DA}$ by double dispatch, meaning that the pair formed by the representations of both operands selects the implementation. Figure 3.5 lists the nine possible combinations.

The nine kernels fall into four families. When both operands are permuted dense, the kernel is a single `dgemm` on the two blocks. When an operand is stacked, the kernel pairs the blocks

Table 3.2 The kernels, the operation each computes, and the atoms of a canonicalized DCP problem that call them, grouped into unary and binary operations. In the second column, $\mathbf{A}$ denotes the Jacobian operand, $\mathbf{B}$ a data matrix, $\mathbf{d}$ a data vector, and $\mathbf{D} = \mathrm{diag}(\mathbf{d})$.

Kernel	Computes	Called by
`transpose`	$\mathbf{A}^\top$	`left_matmul`
`index`	$\mathbf{A}[\mathbf{r}, :]$	`index`, `transpose`
`sum_row_partition`	sums of groups of rows	`sum`
`promote`	copies of a single row	`promote`
`broadcast`	copies of a row or a column	`broadcast`
`diag_vec`	rows moved to the diagonal positions	`diag_vec`
`DA`	$\mathrm{diag}(\mathbf{d})\,\mathbf{A}$	`vector_mult`, `quad_form`
`sum_matrices`	$\mathbf{A} + \mathbf{B}$	`add`, `quad_form`
`BA`	$\mathbf{BA}$	`left_matmul`, `quad_form`
`BTA`, `BTDA`	$\mathbf{B}^\top\mathbf{A}$, $\mathbf{B}^\top\mathbf{DA}$	`quad_form`

of the two operands and calls one `dgemm` for each pair with intersecting row sets. When one operand is dense and the other is generic sparse, the kernel is a hand-written product that multiplies the stored nonzeros of the sparse operand against the dense block, without converting either operand. When both operands are generic sparse, the kernel is Gustavson's algorithm [56]. A product with a permuted dense or stacked operand produces a result of the same kind, so once a dense block appears, the dense representations take over.

The product of two permuted dense matrices stays permuted dense. With the selector matrices of (3.1), the transposed product of $\mathbf{A} = (\mathbf{X}_A, \mathbf{r}_A, \mathbf{c}_A)$ and $\mathbf{B} = (\mathbf{X}_B, \mathbf{r}_B, \mathbf{c}_B)$ expands to

$$\mathbf{B}^\top\mathbf{A} = \mathbf{S}_{c_B}^\top \mathbf{X}_B^\top \underbrace{\left(\mathbf{S}_{r_B}\mathbf{S}_{r_A}^\top\right)}_{\text{index-set intersection}} \mathbf{X}_A \mathbf{S}_{c_A}, \tag{3.2}$$

and $\mathbf{BA}$ expands the same way, with $\mathbf{c}_B$ in place of $\mathbf{r}_B$. The term $\mathbf{S}_{r_B}\mathbf{S}_{r_A}^\top$ keeps the rows that appear in both $\mathbf{r}_B$ and $\mathbf{r}_A$, and the outer selectors place the product at the rows $\mathbf{c}_B$ and the columns $\mathbf{c}_A$, so the result is again one dense block at permuted positions. In practice, the symbolic pass intersects the two sorted index vectors, the numeric pass multiplies the two blocks with one `dgemm`, and no selector matrix is formed. The cost of this operation depends on the sizes of the blocks and not on the global dimensions.

3.3.2 Caching structure across solves

The split between the two passes determines what the engine reuses across solves. In the symbolic pass, a kernel computes everything determined by the structure of its operands and

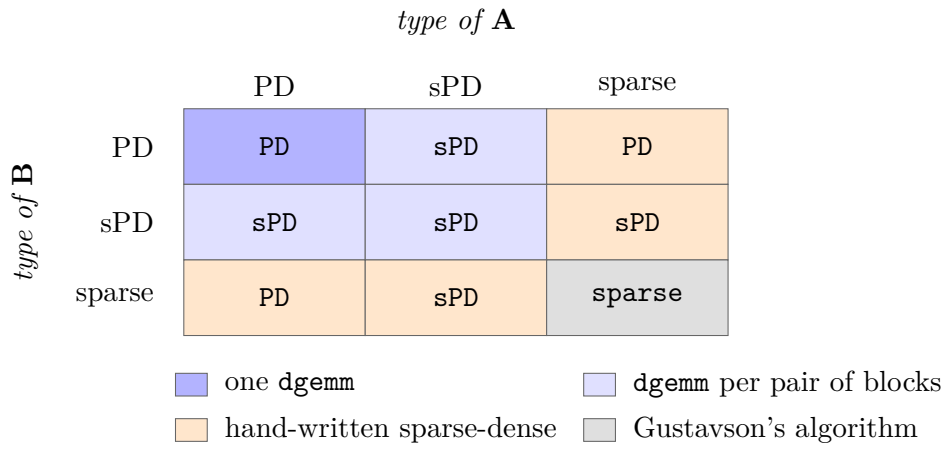


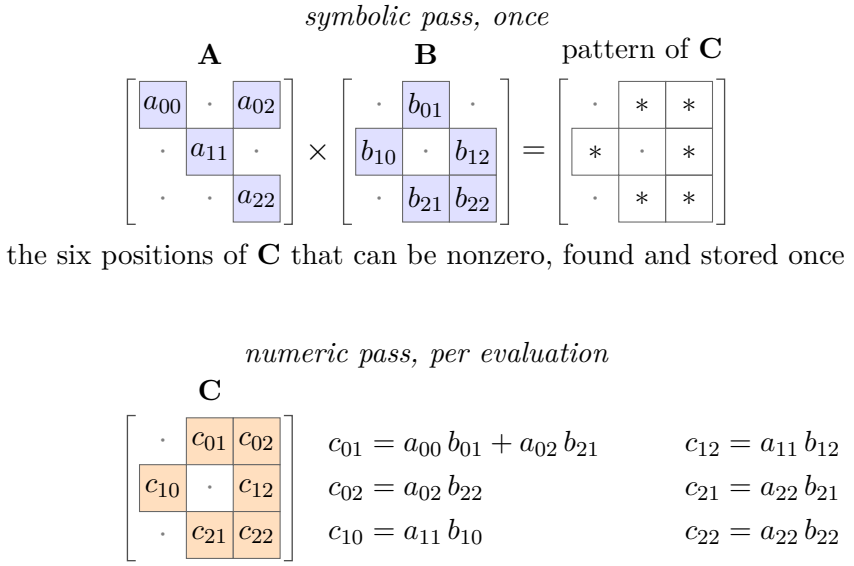
Figure 3.5 Double dispatch of the transposed products $\mathbf{B}^\top\mathbf{A}$ and $\mathbf{B}^\top\mathbf{D}\mathbf{A}$. Each cell names the representation of the result for one combination of operand types (PD is permuted dense and sPD is stacked permuted dense), and its colour indicates how the product is computed.

stores it. In the numeric pass it computes only the values inside that structure, and allocates nothing, because every buffer, including the work arrays of the kernels, is allocated in the symbolic pass. Depending on the kernel, the stored structure is a sparsity pattern, a block layout, or an index map.

The sparse kernels cache sparsity patterns. The `add` atom merges the sorted patterns of its operands, and the sparse products run the symbolic phase of Gustavson's algorithm [56, 59], which finds the pattern of the product row by row. Finding the pattern is the costly and irregular part of both operations, and the numeric pass only computes the values and fills the stored pattern. For a product, the numeric pass computes one sparse dot product per stored nonzero, as shown in Figure 3.6. It does not have to find which entries of the product exist or where they are stored, and evaluates each one directly.

The dense kernels cache block layouts in place of a sparsity pattern. The permuted dense kernels intersect the index vectors of their operands during the symbolic pass, and on the stacked matrices of §3.2.3 the symbolic pass also determines which pairs of blocks have intersecting row sets. The numeric pass gathers the matching rows, multiplies them with `dgemm`, and scatters the products to fixed positions.

The atoms that only move or add entries of the child Jacobian, which we call structural atoms, cache index maps. The Jacobian of `sum` is a sum of groups of rows of the child Jacobian. The symbolic pass records, for each nonzero of the child Jacobian, the output entry it is added to, as shown in Figure 3.7. The numeric pass is one loop with one addition per nonzero. Other structural atoms such as `index`, `transpose`, and `broadcast` build their



the six positions of **C** that can be nonzero, found and stored once

one sparse dot product per stored position, written into place

Figure 3.6 The two passes of a sparse product $\mathbf{C} = \mathbf{AB}$. The symbolic pass finds, from the patterns of **A** and **B**, the positions of **C** that can be nonzero, denoted by $*$, and stores them. The numeric pass computes one sparse dot product per stored position, over the indices at which the row of **A** and the column of **B** are both nonzero.

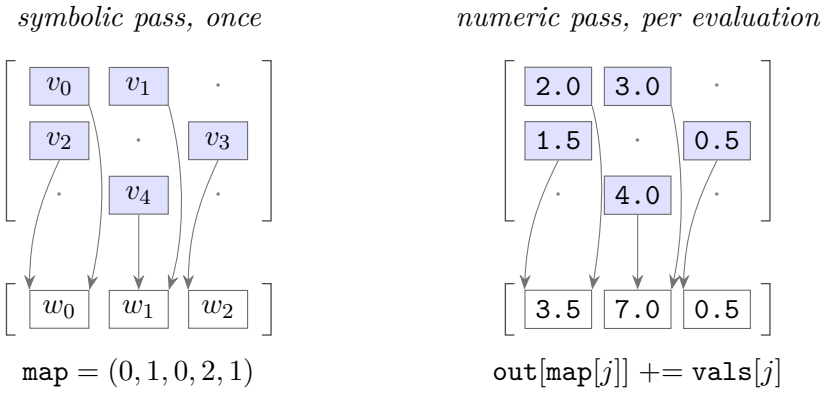


Figure 3.7 The two passes of the `sum` atom for the sum of an expression with three entries. The Jacobian of the sum is the sum of the three rows of the child Jacobian, whose nonzeros v_j are numbered row by row. The symbolic pass computes the pattern of the summed row and the map from each nonzero to the output entry w_i it is added to, shown as arrows. The numeric pass accumulates the values along the fixed arrows.

maps in the same way, with a copy in place of an addition.

3.4 Problem interface

We collect the objective and constraint trees in one `problem` object, defined in `problem.c`. We also wrote `sparsediffpy`, a Python extension module that exposes the atom constructors and the functions that build and evaluate the `problem` object. We keep everything specific to `CVXPY` in a converter and a canonicalization backend named `diffengine`. The lower-case name distinguishes the backend from `SparseDiffEngine`. The converter builds the C trees of the canonicalized problem in one pass over it, with one constructor call per node. The backend obtains the problem data as the values and derivatives of the trees, replacing the matrix stuffing pass of §2.1.

3.4.1 Stacking the constraint Jacobians

The `problem` object stores the constraint trees in the order in which they are registered, and the row blocks of the constraint Jacobian follow that order. Conic solvers expect the constraints grouped and ordered by cone, but the engine knows nothing about cones. Registering the trees in the order the solver expects and recording the cone of each row block are both handled by `CVXPY`, as part of the canonicalization frontend of §1.2.2.

The split between structure and values of §3.3.2 is applied at the problem level as well. The solver receives one global CSR matrix, which the symbolic pass builds once by stacking the sparsity patterns of the constraint Jacobians, as in Figure 3.8. The numeric pass copies the values of each tree into a fixed slice of the global value array. The Jacobian of an affine tree does not depend on the variable values. Its entries are determined by shapes and index sets, or by problem data, and change only when a parameter is updated. The numeric pass therefore skips an affine tree after the first evaluation, until the next parameter update. The Hessian follows the same split. An affine tree contributes nothing to it, so for a canonicalized problem only the objective is evaluated. The Hessian is symmetric, and only its lower triangular entries are passed to the solver.

3.4.2 `CVXPY` interface

The converter maps a variable leaf to its column offset in the stacked variable vector, and a parameter leaf to its offset in the flattened parameter vector $\boldsymbol{\theta}$. Each constraint contributes one tree per argument, in the order in which `CVXPY` groups the cones: zero, nonnegative, second-order, semidefinite, exponential, and power.

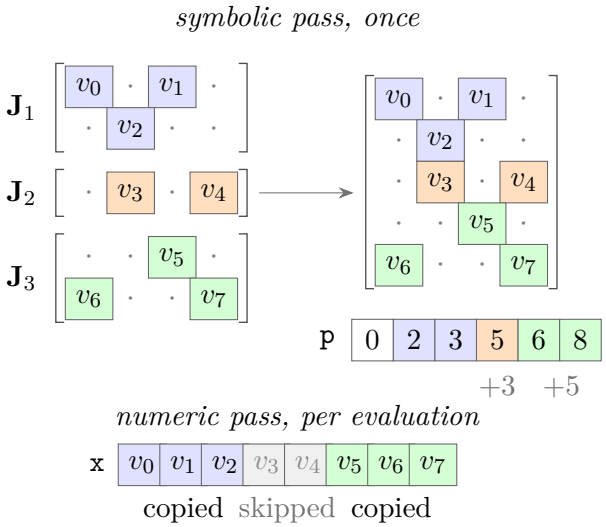


Figure 3.8 The Jacobians of three constraint trees stack into the global constraint Jacobian. The row pointer array `p` concatenates the pointers of the trees, shifted by the nonzeros of the blocks before them. The numeric pass copies the values of each tree into its slice of the value array `x`. The second tree is affine, so the copy is skipped after the first evaluation.

The `diffengine` backend wraps the `problem` object in one class, which has the same interface as the output of matrix stuffing, so the rest of `CVXPY` is unchanged. Listing 3.2 outlines the class.

The class is built once per problem family. Its constructor converts the Python problem to C trees with the converter and runs the symbolic pass. It then evaluates the `problem` object once at the origin, which recovers the problem data of (1.2). `CVXPY` applies the sign convention of the cone form afterwards, by negating $\mathbf{A}$. A parameter update involves only the numeric pass. The method `apply_parameters` writes the new values into the parameter leaves and repeats the evaluations at the origin. After the solve, `split_solution` slices the stacked solution vector into the variables of the original problem. Listing 3.3 shows the evaluations at the origin.

```
class DiffengineConeProgram:
    """minimize (1/2) x'Px + c'x subject to Ax + b in K"""

    C_problem             # the problem object and its buffers
    c, A, b, P            # problem data, updated in place
    constraints, cone_dims # cone info

    def __init__(self, cvxpy_problem): ...
    def apply_parameters(self, id_to_param_value): ...
    def split_solution(self, solution): ...
```

Listing 3.2 A simplified version of the `DiffengineConeProgram` class.

```
problem.init_derivatives()          # symbolic pass, once
A.rows, A.cols = problem.jacobian_sparsity()
P.rows, P.cols = problem.hessian_sparsity()

x0 = np.zeros(n_vars)               # numeric pass, at the origin
problem.objective_forward(x0)       # forward pass, before the gradient
c = problem.gradient()              # objective gradient
b = problem.constraint_forward(x0)  # constraint constants
A.vals = problem.jacobian_values()  # fill the fixed patterns
P.vals = problem.hessian_values()
```

Listing 3.3 A simplified version of the evaluations at the origin that recover the problem data.

3.5 Summary

This chapter presents our approach to canonicalizing optimization problems faster. Our main contribution is two new matrix classes that take advantage of the dense blocks in problem data. Along with these data structures, we present the kernels of Table 3.2 that operate on them. The kernels are new but fairly straightforward, as each reduces to simple index manipulations followed by calls to existing sparse and dense routines. Together, the matrix classes and the kernels let sparse and dense Jacobians mix in the same tree without converting to one specific format.

In addition, for parametric problems, our contribution is the two-phase approach to computing derivatives, which we borrow from the interfaces of NLP solvers outlined in §2.2.3. Although the process is similar, what we store during the symbolic pass is very different.

Instead of a compressed sparsity pattern and a graph colouring, we store index maps and sparsity patterns, and the numeric pass writes the coefficient values directly into them. The cached structure also covers every problem, not only the problems that conform to the DPP ruleset.

Finally, the choice to implement the engine in C with custom matrix classes forced us to think carefully about memory allocation, sparse formats, and opportunities to cache computation. We use the term lightweight to describe the engine because it is a small dependency-free C library (other than BLAS, which is a standard dependency) that only implements the required operations.

The next chapter presents the results of these contributions. For a fair comparison, we compare only against `CVXPY`'s existing backends of §2.1, which take the same canonicalized expressions as input and produce the same problem data. The other modelling languages of Chapter 2 store and compute their coefficients differently. A comparison with them would be difficult to assess, because their times can cover more than just the coefficient computation.

CHAPTER 4 RESULTS

This chapter compares `diffengine`, the canonicalization backend we built in Chapter 3 using `SparseDiffEngine`, against `CVXPY`'s three existing backends. The three are the default `cvxcore`, which we refer to as `CPP`, and the two Python backends `SCIPY` and `COO`, all described in §2.1. Two questions motivate these comparisons. First, we want to evaluate the relative benefit of our proposed symbolic and numeric split against the parameter-to-data tensor. Both approaches store results of the first canonicalization so that subsequent parameter updates are cheap. If the split matches the tensor's re-solve times at a lower upfront cost, building the tensor becomes harder to justify. Second, the `CVXPY` developers have deprecated `cvxcore` and plan to remove it from the library [27]. It nevertheless remains the default backend, in part because neither Python backend has shown a decisive speedup on the first canonicalization. These comparisons allow us to assess whether `diffengine` could be a suitable replacement.

We use the 25 problems of `CVXPY`'s public benchmark suite [86]. The suite includes problems from portfolio selection, regression and classification, image inpainting, and many more applications in convex optimization. On parametric problems, canonicalization follows either the DPP path or the `ignore_dpp` path, both described in §2.1.3. We conduct numerical experiments on both. §4.1 compares the first canonicalization time of all 25 problems on the `ignore_dpp` path. §4.2 compares the parameter-to-data tensor against the symbolic and numeric split on the seven DPP parametric problems of the suite.

All measurements run single-threaded on a 2018 MacBook Pro with a 6-core Intel Core i7-8850H and 16 GB of memory. `CVXPY`'s backends run on the released version 1.9.2. The `diffengine` backend is in a separate branch, available as a pull request:

`https://github.com/cvxpy/cvxpy/pull/3448`

Each measurement runs `get_problem_data` once with `Clarabel` [23] as the target solver, under a time limit of 20 minutes. The method canonicalizes the problem and returns the resulting problem data without calling the solver. We exclude the solver because the choice of backend affects only canonicalization. The benchmark scripts and all recorded timings are available at

`https://github.com/SparseDifferentiation/Jacobian-accumulation`

4.1 First canonicalization

Table 4.1 summarizes the results. `diffengine` is the fastest of the four backends on 21 of the 25 problems, with a geometric-mean ratio of 0.36, and on 17 problems it is more than twice as fast as the best `CVXPY` backend. It is slower on CVaR and OptimalAdvertising, where the times are similar. On SDPSegfault1132 and ConvexPlasticity, processing dense constant data dominates `diffengine`'s canonicalization time.

Figure 4.1a reports the Dolan–Moré performance profile [87] of the canonicalization times. A performance profile compares the backends over the whole suite at once. We define $t_{p,s}$ as the time of backend s on problem p. The profile of backend s is the fraction of the n problems on which it is within a factor τ of the fastest backend,

$$f_s(\tau) = \frac{1}{n} \left| \{\, p : t_{p,s} \le \tau \, \min_b t_{p,b} \,\} \right|. \tag{4.1}$$

The minimum $\min_b t_{p,b}$ is the time of the fastest backend on problem p. The value at $\tau = 1$ is the fraction of problems on which the backend is the fastest, and the curve reaches 1 at the factor the backend needs to cover every problem. `diffengine` is within $3\times$ of the best backend on every problem. No `CVXPY` backend is fastest on more than two problems, and each needs $\tau \approx 8$ to cover 92–96% of the problems.

Figure 4.1b reports the same profile for peak memory, with $t_{p,s}$ replaced by the growth of the process's peak resident memory during canonicalization. `SCIPY` has the lowest peak memory on 16 of the 25 problems. `diffengine` matches `CPP` and `COO` in geometric mean and uses about $1.4\times$ the memory of `SCIPY`.

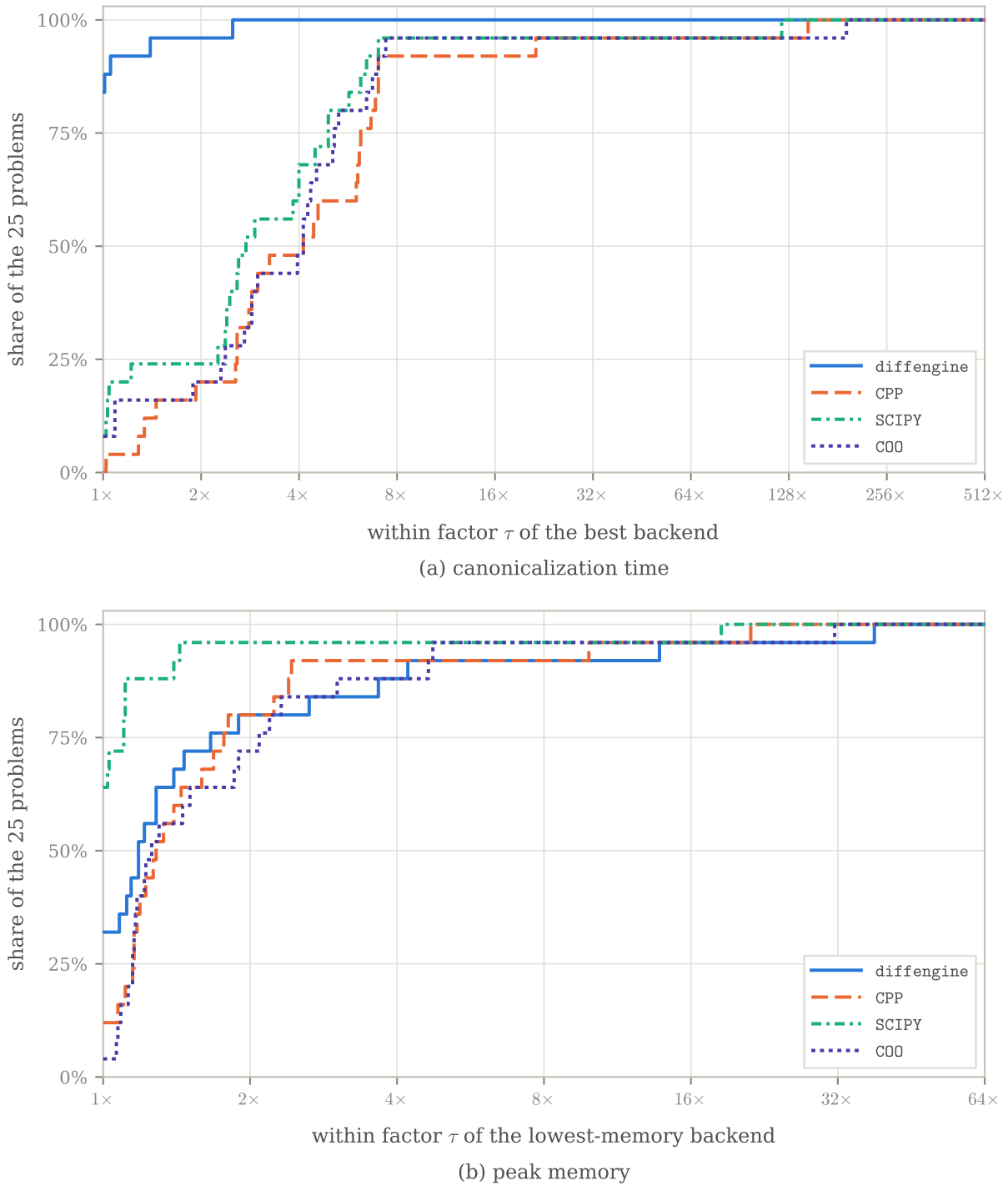


Figure 4.1 Performance profiles (4.1) of canonicalization time (a) and peak memory (b) over all 25 problems, every backend on the `ignore_dpp` path.

Table 4.1 `ignore_dpp` canonicalization time in seconds on all 25 problems. Bold denotes the fastest of the four backends on each problem. The ratio is `diffengine` over the best of `CVXPY`'s three backends. The final row is the geometric mean of the per-problem ratios.

Benchmark	`CPP` (s)	`SCIPY` (s)	`COO` (s)	`diffengine` (s)	Ratio
UnconstrainedQP	10.32	8.52	13.54	**0.07**	0.01
LeastSquares	9.61	10.08	10.68	**1.45**	0.15
Yitzhaki	3.56	3.28	3.55	**0.51**	0.16
HuberRegression	12.05	10.72	11.60	**1.73**	0.16
SimpleQP	11.27	10.59	11.97	**1.87**	0.18
FactorCovarianceModel	5.26	4.22	4.40	**0.86**	0.20
SimpleLP	37.57	29.99	30.96	**6.12**	0.20
SVMWithL1Regularization	11.14	8.24	9.78	**1.86**	0.23
SimpleScalarParametrizedLP	4.54	4.08	4.20	**1.03**	0.25
SimpleFullyParametrizedLP	2.22	2.16	2.22	**0.54**	0.25
ParametrizedQP	11.51	9.68	10.76	**2.53**	0.26
TvInpainting	3.89	3.50	3.57	**1.20**	0.34
Cajas	22.67	2.91	2.89	**1.07**	0.37
SlowPruning	7.69	7.82	8.60	**3.01**	0.39
SmallMatrixStuffing	6.97	7.09	10.84	**2.76**	0.40
QuantumHilbertMatrix	4.27	4.05	4.75	**1.67**	0.41
ConeMatrixStuffing	11.06	9.34	16.99	**3.95**	0.42
ParamSmallMatrixStuffing	0.28	0.50	0.49	**0.21**	0.75
ParamConeMatrixStuffing	0.18	0.32	0.32	**0.14**	0.79
SemidefiniteProgramming	5.96	2.43	3.78	**2.02**	0.83
Murray	7.13	3.87	4.00	**3.72**	0.96
CVaR	35.85	**24.79**	26.86	24.94	1.01
OptimalAdvertising	6.43	2.32	**2.25**	2.36	1.05
SDPSegfault1132	64.84	**9.51**	43.00	13.21	1.39
ConvexPlasticity	**0.35**	**0.35**	**0.35**	0.85	2.48
Geometric mean					**0.36**

4.2 Parametric canonicalization

Table 4.2 reports first canonicalization and re-solve times on the seven DPP parametric problems. We refer to a re-solve in this context as rebuilding the problem data after a parameter update, so the reported times again exclude the solver. On the first canonicalization, each `CVXPY` backend runs the DPP path and builds a parameter-to-data tensor. `diffengine` instead runs its symbolic pass and one numeric pass. It has the fastest first canonicalization on every problem, and among the `CVXPY` backends only `COO` is able to build the tensor on all seven. On a parameter update, the three `CVXPY` backends re-solve by applying the cached tensor, so the tensor column is common to all three. `diffengine` instead runs another numeric

pass.

Table 4.2 First canonicalization and re-solve time in seconds on the seven DPP parametric problems. A dash denotes a tensor that did not finish building within the time limit. Bold denotes the fastest first canonicalization and the faster re-solve. The ratio is the re-solve time of `diffengine` over the re-solve time of the tensor.

	first canonicalization				re-solve		
Benchmark	`CPP`	`SCIPY`	`COO`	`diffengine`	tensor	`diffengine`	Ratio
ParamSmallMatrixStuffing	7.78	29.06	0.40	**0.21**	0.005	**0.001**	0.29
ParamConeMatrixStuffing	3.16	7.63	0.26	**0.14**	0.003	**0.001**	0.29
SimpleFullyParametrizedLP	—	—	2.17	**0.54**	0.261	**0.126**	0.48
ParametrizedQP	—	—	10.87	**2.53**	1.485	**0.985**	0.66
SimpleScalarParametrizedLP	4.68	4.15	4.48	**1.03**	0.304	**0.247**	0.81
SVMWithL1Regularization	11.00	8.21	9.62	**1.86**	**0.252**	0.496	1.97
FactorCovarianceModel	4.92	4.15	4.66	**0.86**	**0.139**	0.279	2.00

On SVMWithL1Regularization and FactorCovarianceModel, the two problems where the parameter map re-solves faster, the problem data is almost entirely constant and the parameters are a few scalars. On the SVM problem the parameter is a single regularization weight that touches one coefficient of the problem data. A parameter update nevertheless forces the numeric pass to recompute the entire constant part, which can be costly.

4.3 Discussion

The benchmark experiments in this chapter lead us to make the following three observations:

- Caching canonicalization does not necessarily require the DPP ruleset. In Table 4.2, the symbolic and numeric split re-solves within a factor of about two of the cached tensor on every problem while ignoring the ruleset.
- The tensor approach carries a large upfront cost that is not always justified. In Table 4.2, the split's first canonicalization is faster than building the tensor on every problem, and its re-solve is also faster on five of the seven problems.
- `cvxcore` can be replaced. `diffengine` canonicalizes faster on 24 of the 25 problems, as reported in Table 4.1 and Figure 4.1. On the parametric problems of Table 4.2, `COO` builds the parameter-to-data tensor faster than `cvxcore` on all seven.

Traditionally, `CVXPY` has relied on DPP to guarantee a meaningful speedup between the first and subsequent canonicalizations of a parametric problem. If a problem does not conform

to the DPP grammar, users have to either attempt one of many possible reformulations or accept that every subsequent canonicalization will cost the same as the first. In addition, even if the problem conforms to the grammar, forming the tensor is sometimes prohibitive, especially for large-scale problems. Our first two observations challenge this tradition and suggest that DPP should be used more carefully, for problems that are re-solved enough times or for applications built on downstream packages that depend on the tensor, for example `CVXPYgen` [66] and `CVXPYlayers` [26]. With our approach, a similar speedup can be obtained through the symbolic and numeric split without enforcing any rules about how parameters appear in the problem formulation.

In addition, the second and third observations suggest that `diffengine` can replace `cvxcore` as the default backend of `CVXPY`. Canonicalization is faster on nearly every benchmark problem, on both the first and subsequent canonicalizations, so there is little reason to use `cvxcore` in the future. Making this change will require substantial work and coordination with the maintainers of `CVXPY`. We discuss these next steps and the roadmap of future work in §5.2. Finally, `diffengine` has downsides. Its peak memory is higher on several problems, and its re-solve requires the cached program to stay in memory. The benchmark suite also covers only a subset of possible optimization problems, and more experiments would be needed to validate these observations further. We return to these limitations in §5.1.

CHAPTER 5 CONCLUSION

Throughout this Master's thesis, we revisit the performance of compiling a high-level model into conic problem data. Despite being essential to solving optimization problems, the compilation step has mostly been an afterthought. The reason for this is that the cost is often negligible, and when it is not, most users accept it as necessary. In our work, we take the perspective of computing the coefficients of the problem data as sparse derivatives, which differs from most modelling languages. The closest is `CasADi`, whose algorithmic differentiation (AD) is built for general nonlinear functions. Applied to affine functions, where the coefficients follow directly from the constants of the expression graph, `CasADi` would still need to detect the sparsity patterns at runtime and compress the Jacobian computation by graph colouring. `AMPL` and `JuMP` take the coefficients as the user declares them, but both scalarize the model. A dense block in the problem data becomes a list of scalar coefficients. `CVXPY`'s backends, on the other hand, convert everything to sparse matrices. No existing modelling language keeps dense blocks as they are. To address this, we built `SparseDiffEngine`, a sparse differentiation library that operates on vectorized expressions without scalarizing them, requires no sparsity detection or graph colouring, and mixes sparse and dense formats throughout the computation. In addition to using BLAS operations and dense storage on dense blocks, we implement a two-phase approach which caches many computations during the symbolic pass. For parametric problems, the numeric pass reuses these computations, with no restrictions on how the parameters enter the problem.

We also propose `diffengine`, a new canonicalization backend for `CVXPY` with a simple interface to `sparsediffpy`, the Python bindings of `SparseDiffEngine`. We compare `diffengine` with `CVXPY`'s three existing backends in numerical experiments and obtain promising results. They allow us to conclude that `diffengine` can replace `cvxcore` as the default backend of `CVXPY` in certain circumstances. Overall, the work in this thesis is based on a combination of fundamental subjects in matrix computations, computing derivatives, and modelling languages for optimization.

5.1 Limitations

The benchmark suite of `CVXPY` is a collection of synthetic problems accumulated over the years by the library's developers [86]. However, it is not a standard benchmark collection like the ones maintained for optimization solvers. Two examples are the Mittelmann benchmarks [88] and the repository of convex quadratic programs (QPs) of Maros and Mészáros [89]. Un-

fortunately, these collections distribute their instances in standard form, ready to be passed to the solver, so there is no high-level model to compile. Our conclusions therefore rely on a single suite, and their generality is not yet completely validated. If and when a standard benchmark suite written at the modelling level becomes available, the experiments of Chapter 4 should be run again on it. It would also allow us to begin comparing compilation times across modelling languages. The difficulty in creating such a suite is that every modelling language has its own syntax and conventions, so the problems would have to be written once per language.

The results of Chapter 4 also show that the peak memory of `diffengine` is higher on several problems of the suite. Part of the cost is the cached problem itself. The numeric pass evaluates the expression trees at every parameter change, so the `problem` object must stay in memory for the lifetime of the problem family. The parameter-to-data tensor does not have this constraint. Once the tensor is built, a re-solve needs only its product with the current parameter values, and the formulation can be set aside. This decoupling brings similar benefits as `AMPL`'s `.nl` format, which separates the problem instance from the model. The separated data can be archived, shipped to another machine, and used by downstream packages such as `CVXPYgen` [66] and `CVXPYlayers` [26].

5.2 Future Research

The contributions of this thesis range from the creation of new packages, to improvements of existing ones, to new data structures and algorithms. For this reason, we see three directions for future work, one in each type of contribution. The first direction concerns `sparsediffpy`. We plan to add a standalone Python interface, so that the sparse Jacobians and Hessians can be built and evaluated without `CVXPY`. We believe there can be many applications of such derivatives without necessarily calling an optimization solver. The user would build the expression trees directly with the functions of `sparsediffpy` instead of with `CVXPY` expressions.

The second direction is closer to development than research. It covers the remaining work on the `diffengine` backend within `CVXPY`. The immediate goal is to make `diffengine` the default backend on the `ignore_dpp` path. One challenge is that existing reductions must change behaviour without breaking backwards compatibility. The next task is the removal of `cvxcore`, which the `CVXPY` developers have already deprecated. The removal would mean that `CVXPY` no longer needs to compile and ship C++ binaries, which would simplify the build process of the library. As a replacement, `diffengine` can become the new default backend, with `COO` as an alternative candidate for disciplined parameterized programming

(DPP) problems.

The final direction would be to extend the permuted dense matrix classes of Chapter 3 to the rest of the optimization process. Both the presolve and the solver operate on a generic sparse matrix, where the dense blocks are no longer explicit. The high-level step would be to pass the block structure along with the problem data, so that the presolve and the factorization can work on the dense blocks directly instead of rediscovering them in a sparse matrix [57]. This change would require much more research, because nothing beyond basic intuition indicates that it is worth doing. For existing presolve routines and sparse linear system solvers, it would also require adapting their code to operate on matrix representations other than the typical sparse matrix.

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